

Stronger symmetric projection stability

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7 October 2026

This complete proof collection establishes the two globally valid bounds $D(K, \mathcal{E}_d) - 1 \leq \min\{d^{12}\delta^{1/(3d)}, d^{19}\delta^{1/[3(d-1)]}\}$ for origin-symmetric full-dimensional bodies in dimension $d \geq 3$. The optimum remains unresolved. Section and equation labels below are local to each named component. The first component states all assumptions and constants; the second supplies the retained cone-law input and whole-class obstruction. The historical exponent and gap statements in Component II refer to the preserved baseline theorem; Component I gives the stronger current modulus and narrower unresolved gap. No quantitative inverse-Minkowski theorem is used.

I Main theorem and assembly

A stronger upper-end exponent from weighted rank-two matching

7 October 2026. This note proves a strengthening of the completed symmetric projection-cone stability theorem. The completed supplement is unchanged. The new steps are a sharp square-root matching estimate and a direct recovery theorem from off-block normals. No heuristic tuning of thresholds is used.

1. Statement

Let $K \subset \mathbb{R}^d$ be a centrally symmetric full-dimensional convex body, centered at zero, and let $a(K)$ be the entrywise affine projection-cone invariant. Set $\delta = 1/2 - a(K)$. Let $\mathcal{E}_d$ be the entire class of linear images of Cartesian products of centrally symmetric convex factors of dimensions one and two. Use

$$d_{\text{BM}}(K, E) = \inf\{\lambda \geq 1 : TE \subseteq K \subseteq \lambda TE, T \in \text{GL}(d)\}, \quad D(K, \mathcal{E}_d) = \inf_{E \in \mathcal{E}_d} d_{\text{BM}}(K, E).$$

Theorem 1 (weighted matching). For every $d \geq 3$ and every such K ,

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq d^{12} \left(\frac{1}{2} - a(K)\right)^{1/(3d)}}. \quad (1)$$

There is no smoothness, curvature, polytope, atom, or absolute-continuity hypothesis. In dimensions one and two the distance is one.

For the explicit unsimplified constants put

$$\begin{aligned} N &= d + 1, & \kappa_d &= \pi^{d/2}/\Gamma(1 + d/2), & a_d &= d^{-2d}, \\ Z_d &= 2^{d+9} N \sqrt{d} \binom{d}{3} / \kappa_d, & F_d &= 13N + 3Z_d, & B_d &= 4NF_d/a_d, \\ \hat{L}_d &= 5d^2 a_d^{-2} B_d^{1/2}, & \hat{H}_d &= d(\sqrt{d} + 1)\hat{L}_d, & \hat{C}_d &= 2d(2d\hat{H}_d)^{1/d}. \end{aligned}$$

The proof gives the stronger constant $\hat{C}_d \leq d^{12}$, and

$$D(K, \mathcal{E}_d) - 1 \leq \hat{C}_d \delta^{1/(3d)}. \quad (2)$$

If

$$0 < \delta \leq \hat{\delta}_d^* := \left(\frac{2^{-d}}{2d\hat{H}_d} \right)^3,$$

there is also the explicit containment estimate

$$D(K, \mathcal{E}_d) \leq \frac{1}{1 - (2d\hat{H}_d \delta^{1/3})^{1/d}}. \quad (3)$$

The constants are conservative and depend on dimension. The earlier all-class corner-truncation obstruction still excludes exponents above $1/d$. The next theorem further improves the geometric exponent.

Theorem 2 (weighted matching and direct geometric recovery). Under exactly the same hypotheses,

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq d^{19} \left(\frac{1}{2} - a(K)\right)^{1/[3(d-1)]}}. \quad (8)$$

Both estimates hold globally, so their minimum is also a valid modulus. The remaining optimal-exponent interval is $[1/[3(d-1)], 1/d]$. The factor d^{19} is deliberately coarse.

2. The inputs retained from the completed proof

The normalized cone law is

$$\nu_K = (u \mapsto u/h_K(u))_* \left[\frac{h_K(u)}{d|K|} dS_K(u) \right].$$

For independent samples X_i define

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E} \left| \det \begin{pmatrix} X_1 & \cdots & X_{d+1} \\ 1 & \cdots & 1 \end{pmatrix} \right|.$$

Entry005 v3 proves $a(K) = B/((d+1)A)$ for general convex bodies. For a $(d+1)$ -tuple, write ϕ for its fourth-largest absolute horizontal cofactor. The completed proof supplies the following three established inputs.

1. In symmetric John position $B_2^d \subseteq K \subseteq \sqrt{d}B_2^d$, the law is supported in the unit ball and $A \geq d^{-3d/2} \geq a_d$. This follows from the regular-contact identity $\mathbb{E}[YX^\top] = I/d$ and sequential wedge expectations.
2. The cone-specific contact-slab argument gives $\mathbb{E}\phi \leq F_d \delta^{2/3}$. Its conditional determinant denominator cancels before integration. The estimate includes singular tuples, nonatomic laws, flat-face atoms, and the zero-deficit case.
3. If Q is centered symmetric and full-dimensional, $Q \subseteq RB_2^d$, and $\mathcal{W}$ is the union of an orthogonal block decomposition, then its product of block projections P satisfies

$$(1 - (2dv)^{1/d})P \subseteq Q \subseteq P, \quad v = R(\sqrt{m} + 1)\mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W}), \quad 2dv < 1. \quad (4)$$

Here m is the number of blocks. The proof bounds $V(Q[d-1], P)/|Q| - 1$ by v , uses Minkowski's first inequality, and inserts a homothetic image of a symmetric half of P into its missing cap. It assumes no inradius bound or inverse cone-volume uniqueness theorem.

These inputs are Sections 2, 3 and 5 of the completed English proof. For Theorem 1 the replaced argument is its Section 4 matching estimate; Theorem 2 also replaces the final geometric recovery step. The full new matching proof is in the accompanying [matching.md](#), and its essential quantitative mechanism is recorded next.

3. Why weighted matching improves the power

For a bounded Borel law with $A \geq a > 0$, $\mathbb{E}\phi \leq \varepsilon$, put $q = 4(d+1)\varepsilon/a$. The simultaneous Fubini-section selection in [matching.md](#) gives a sampled basis U , with $D = |\det U| \geq a/2$, whose one-sample and every omitted-basis two-sample section each have fourth-cofactor expectation at most q . In particular

$$\|U\| \leq \sqrt{d}, \quad \|U^{-1}\| \leq 2\sqrt{d}/a, \quad |\alpha_i| \leq 1/D, \quad \mathbb{E}s_3(\alpha) \leq q/D^2, \quad \alpha = U^{-1}X,$$

where s_3 is the third-largest coordinate magnitude.

Retain the two largest coordinates of each vector, with fixed-index tie breaking. In an omitted-basis section the cofactors, up to signs, are

$$D\alpha_j, \quad D\beta_j, \quad D(\alpha_i\beta_j - \alpha_j\beta_i) \quad (i \neq j).$$

Truncating both vectors changes each entry by at most $2s_3(\alpha) + 2s_3(\beta)$. Absolute value and a fixed order statistic are 1-Lipschitz in the maximum norm, so the truncated section expectation is at most $q + 4q/D^2$.

Give a coordinate pair ij the weighted mass

$$m_{ij} = \mathbb{E} \left[\min(|\alpha_i|, |\alpha_j|) \mathbf{1}_{\{ij \text{ is the chosen top pair}\}} \right].$$

For adjacent pairs ij, jk , their two truncated vectors make four cofactors at least $D^2 r_\alpha r_\beta$, where r denotes the smaller retained magnitude. Thus

$$m_{ij} m_{jk} \leq \frac{q + 4q/D^2}{D^2} \leq \frac{5q}{D^4}. \quad (5)$$

Pairs with mass exceeding $\sqrt{q + 4q/D^2}/D$ form a matching. Use them as plane blocks and the remaining indices as line blocks. A discarded pair contributes its smaller coordinate to distance from the chosen block union, rather than its entire probability mass. Consequently, for $0 \leq q \leq 1$,

$$\begin{aligned} \mathbb{E} \text{dist}(\alpha, \mathcal{W}) &\leq \frac{\sqrt{d}q}{D^2} + \binom{d}{2} \frac{\sqrt{q + 4q/D^2}}{D} \\ &\leq \frac{\sqrt{d} + \sqrt{5}\binom{d}{2}}{D^2} \sqrt{q} \leq 5d^2 a^{-2} \sqrt{q}. \end{aligned} \quad (6)$$

The last inequality follows from $4(\sqrt{d} + \sqrt{5}\binom{d}{2}) \leq 4d + 5d(d-1) \leq 5d^2$. For $q = 0$, all positive-weight edges form a matching and the third coordinate vanishes almost surely. This proves the zero case without a limit. All selections are finite Borel partitions, so no point needs positive mass.

4. Complete assembly of the stronger theorem

Put K in symmetric John position, and apply (6) with $a = a_d$, $\varepsilon = F_d \delta^{2/3}$. Then

$$q = B_d \delta^{2/3}, \quad \mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) \leq \hat{L}_d \delta^{1/3}. \quad (7)$$

For $\delta \leq \hat{\delta}_d^*$, the condition $q \leq 1$ is valid: $\hat{H}_d \geq B_d^{1/2}$, and therefore

$$q \leq B_d \left(\frac{2^{-d}}{2d\hat{H}_d} \right)^2 \leq 1.$$

Apply the primal map $U^\top$. The body $Q = U^\top K$ is contained in dB_2^d ; its cone law is exactly $(U^{-1})_* \nu_K$. Its coordinate line and plane blocks are orthogonal. Thus (4), with $R = d$, $m \leq d$, and (7), gives (3). Every projection factor is centrally symmetric and full-dimensional in its block, so $P \in \mathcal{E}_d$. No decomposition has to be canonical or unique.

At the smallness threshold,

$$r := (2d\hat{H}_d \delta^{1/3})^{1/d} \leq 1/2.$$

The sandwich $(1-r)P \subseteq Q \subseteq P$ gives

$$D(K, \mathcal{E}_d) - 1 \leq \frac{r}{1-r} \leq 2(2d\hat{H}_d)^{1/d} \delta^{1/(3d)}.$$

For every larger deficit use the John sandwich with a cube, which gives $D(K, \mathcal{E}_d) \leq d$. The global coefficient in (2) covers this range because

$$\hat{C}_d(\hat{\delta}_d^*)^{1/(3d)} = d.$$

At $\delta = 0$, (6) gives exact block support and (4) gives the product itself. For $d = 1, 2$, K is already an allowed factor. This proves (2) and (3) in all cases.

To prove the stated numerical constant, the inscribed cube bound $\kappa_d \geq (2/\sqrt{d})^d$ yields

$$Z_d \leq 2^{10} d^{d/2+9/2}, \quad F_d \leq 2^{12} d^{d/2+9/2}, \quad B_d \leq 2^{15} d^{(5/2)d+11/2}.$$

Consequently

$$\hat{L}_d \leq 5 \cdot 2^{15/2} d^{(21/4)d+19/4}, \quad \hat{H}_d \leq 10 \cdot 2^{15/2} d^{(21/4)d+25/4},$$

and

$$\hat{C}_d^d \leq 20 \cdot 2^{d+15/2} d^{(25/4)d+29/4}.$$

For $d \geq 3$, $20 \cdot 2^{15/2} < 4096 \leq d^8$, while $2^d \leq 3^{d-1} \leq d^{d-1}$: the first inequality starts at $8 \leq 9$ and its ratio decreases by $2/3$ with each increment of d . Hence

$$\hat{C}_d^d \leq d^{(29/4)d+57/4} \leq d^{12d},$$

where the last inequality is exactly $19d \geq 57$. This proves (1). $\square$

4.1. The direct geometric recovery theorem The full proof in [geometry-rigidity.md](#) gives the following stronger geometric input. For a centered symmetric full-dimensional body $Q \subset RB_2^d$, an orthogonal decomposition into m nonzero blocks, its projection product P , and $s = \mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W})$, put

$$T = (2^{2d-1} d R s)^{1/(d-1)}.$$

Then

$$(1 - mT)P \subset Q \subset P \tag{9}$$

whenever $mT < 1$. In particular the Banach–Mazur excess to this product is at most $2mT$ when $mT \leq 1/2$. If the factors have dimensions one or two, the globally valid whole-class excess is at most $2mdT$. No inradius or inverse-basis norm occurs in these estimates.

For clarity, the mechanism and constants can be summarized independently. For an orthogonal bipartition $E \oplus F$, write $P_{E,F}$ for the two-factor projection product, $\lambda P_{E,F} \subset Q$ for its maximal centered homothetic inclusion, and $\tau = 1 - \lambda$. A support witness u, p has $u \cdot p = H = h_{P_{E,F}}(u)$ and $h_Q(u) = (1 - \tau)H$. Lift the two block components of p to $q_A, q_B \in Q$. Their differences from p span a two-dimensional plane L with orthogonal axes in E, F . Its normal component u_L is nonzero.

Contract p, q_A, q_B by $1 - \eta$ toward any interior point of Q , with $\eta = \tau/4$. The contracted witness remains outside Q by support height at least $\tau H/2$. In every resulting planar section, the mixed-normal boundary length is therefore at least $\tau H/(2|u_L|)$. The planar assertion is proved by writing its negative quadrant frontier as $y = -g(x)$: convexity makes g' nondecreasing, and $\int \min(1, g') = \min(g(x) - x)$ is at least the distance to the witness.

The transverse parameters fill a scaled copy of $\text{proj}_{L^\perp} Q$ of measure $\eta^{d-2} |\text{proj}_{L^\perp} Q|$. The coarea Jacobian on the regular boundary is $|\text{proj}_L n|$, so

$$I_{E,F} := \int_{\partial Q} \min(|\text{proj}_E n|, |\text{proj}_F n|) d\mathcal{H}^{d-1} \geq \eta^{d-2} |\text{proj}_{L^\perp} Q| \frac{\tau H}{2|u_L|}.$$

Every planar fiber has area at most $4RH/|u_L|$, by the same support slab and the outer-radius bound. Fubini cancels the transverse projection and support denominator, giving

$$\tau^{d-1} \leq 2^{2d-1} R I_{E,F} / |Q|.$$

For a single-block bipartition, its normal integrand is at most $\text{dist}(n, \mathcal{W})$. Homogeneity of distance cancels the cone weight, so the latter integral is exactly $d|Q|s$, and every single-block homothetic loss is at most T .

The single-block inclusions multiply: successively replace coordinates by their desired projection-product values, scaled by the preceding inclusion factors. The resulting point is in Q and has all coordinates scaled by $\prod_j \lambda_j$. Thus $(\prod_j \lambda_j)P \subset Q$, with $\prod_j \lambda_j \geq 1 - \sum_j (1 - \lambda_j) \geq 1 - mT$, proving (9). The full note supplies the planar lemma and coarea regularity details for arbitrary convex boundaries, including flat facets and nonatomic laws.

4.2. Complete assembly and numerical constant for Theorem 2 Retain $M_d = \hat{L}_d = 5d^2 a_d^{-2} \sqrt{B_d}$, and define

$$G_d = (2^{2d-1} d^2 M_d)^{1/(d-1)}, \quad C'_d = 2d^2 G_d, \quad \beta_d = \frac{1}{3(d-1)}.$$

The coordinate body from (7) has $R = d$, $m \leq d$, and $s \leq M_d \delta^{1/3}$. Therefore $T \leq G_d \delta^{\beta_d}$. For

$$0 < \delta \leq \delta'_d := (1/(2dG_d))^{3(d-1)},$$

we have $mT \leq 1/2$. The matching condition $q \leq 1$ also holds: $M_d \geq \sqrt{B_d}$, hence $G_d^{d-1} \geq \sqrt{B_d}$, and

$$q \leq B_d / (2dG_d)^{2(d-1)} \leq 1.$$

The small-deficit excess is consequently at most $2dG_d \delta^{\beta_d}$. At all larger deficits use the John bound $D - 1 \leq d - 1$; it is covered by the global coefficient since $C'_d (\delta'_d)^{\beta_d} = d$. At zero deficit, the exact matching case and (9) give the product itself. This proves $D - 1 \leq C'_d \delta^{\beta_d}$ in every case.

To bound C'_d , the earlier estimate on B_d gives

$$M_d \leq 5 \cdot 2^{15/2} d^{(21/4)d+19/4}, \quad (C'_d)^{d-1} \leq 5 \cdot 2^{3d+11/2} d^{(29/4)d+19/4}.$$

For $d \geq 3$, $8 \leq d^2$, and $5 \cdot 2^{11/2} < 243 \leq d^5$. Thus

$$(C'_d)^{d-1} \leq d^{(37/4)d+39/4} \leq d^{19(d-1)},$$

where the final inequality is exactly $39d \geq 115$, valid already at $d = 3$. Hence $C'_d \leq d^{19}$, proving (8). $\square$

5. What is genuinely sharp, and what remains open

The accompanying complete notes isolate different hypotheses.

- [halfmass.md](#) proves that the cone-specific power $2/3$, its near-half-mass tail power two, and the contact-slab area power two are sharp for actual cone laws in every fixed $d \geq 3$. The bodies are a doubly truncated three-cube times intervals, already in John position.
- [matching.md](#) proves that the improved power $1/2$ is sharp for bounded laws with determinant mass bounded below, even over every choice of basis and every line/plane decomposition. Its first obstruction is an arbitrary even law. The separate [global-next.md](#) strengthens the sharpness statement to actual cone laws and proves a geometric all-product distance obstruction for that family.

- [geometry-next.md](#) proves that the cap exponent $1/d$ is sharp under volume-only information. With the stronger off-block normal measurement, truncated cubes instead exclude powers above $1/(d-1)$. The general recovery theorem with power $1/(d-1)$, now proved directly in [geometry-rigidity.md](#), closes this intermediate geometric gap. It is sharp for the stronger normal input.

These sharpness examples do not establish optimality of (8): independent losses need not be simultaneously sharp on the same cone law. They rule out improving the first two isolated lemmas under their present hypotheses, and explain why a further gain should retain more geometry or couple the stages.

The universally valid weighted relation

$$\mathbb{E}[\phi g] \leq (d+1)A\delta, \quad g = 1/2 - \max_i (|c_i| / \sum_j |c_j|),$$

with $g = 0$ on the singular event $\sum_j |c_j| = 0$, retains information lost by the unweighted fourth-cofactor moment. On the sharp half-mass family, the actual mixed-volume support surplus is linear in δ . However, inserting an extra global containment factor into the support-surplus estimate is false: the explicit near-coordinate cuts in [geometry-next.md](#) give a counterexample. No unproved coupled support-surplus theorem is used in either bound.

No quantitative inverse-Minkowski theorem is a proof input. The direct section/coarea proof supplies the geometric estimate used here.

6. Dependency and verification scope

The complete baseline proof is included in [dependencies/completed-proof.md](#). Its corrected text has SHA-256 6a31c8d2dff76b3f73b862a0b6ef2ea192afc9de23b3605208b465ee3a5c1193; the immutable original has SHA-256 08f6f55c6a694da05d348d3dcbda18be186044fd39570c51bf786507aaef862c. The baseline pins the v2–v4 sources of `mxym/math` to 6785c1c830f8e19e2eb07b0bb89f4d475a8b154a. The exact dependency map remains applicable. Bibliographic attributions and comparisons are corrected separately; this package includes the corrected baseline mathematical proof in [dependencies/completed-proof.md](#). Neither the lower-end simplex modulus nor the v5 spectral recursion is imported here.

The new rational programs check the polynomial identities and exact examples described in their adjacent notes. They use explicit exceptions and retain checks under Python optimization. They do not prove the general integral, surface-measure, or convex-geometric assertions. The written proof and its independent model audit are separate from these finite regressions. No human peer review, proof-assistant verification, or novelty claim is made.

The optimal final exponent remains unresolved between $1/[3(d-1)]$ and $1/d$. The isolated intermediate examples do not establish optimality of the composed bound. The original completed release is preserved.

II Retained baseline proof and whole class obstruction

1. The theorem and explicit constants

Let $K \subset \mathbb{R}^d$ be any full-dimensional centrally symmetric convex body, with its center translated to zero. There is no polytope, smoothness, strict-convexity, absolute-continuity, or positive-curvature assumption. Retain the entry005 invariant

$$a(K) = \left(\frac{d}{d+1} \right)^d \frac{R(\mathcal{P}K)}{R(K)} - 1, \quad R(K) = \frac{|\Pi K|}{|K|^{d-1}}.$$

Let $\mathcal{E}_d$ consist of all linear images of Cartesian products of centrally symmetric factors of dimensions one and two. For centered symmetric bodies use

$$d_{\text{BM}}(K, L) = \inf\{\lambda \geq 1 : TL \subseteq K \subseteq \lambda TL, T \in \text{GL}(d)\}, \quad D(K, \mathcal{E}_d) = \inf_{E \in \mathcal{E}_d} d_{\text{BM}}(K, E).$$

Write $\delta = \frac{1}{2} - a(K)$, which is nonnegative by v3.

Theorem A. Explicit power stability for the entire equality class. For $d = 1, 2$, $D(K, \mathcal{E}_d) = 1$. For every $d \geq 3$, define

$$\begin{aligned} N &= d+1, & \kappa_d &= \frac{\pi^{d/2}}{\Gamma(1+d/2)}, & a_d &= d^{-2d}, \\ Z_d &= \frac{2^{d+9} N \sqrt{d} \binom{d}{3}}{\kappa_d}, & F_d &= 13N + 3Z_d, & B_d &= \frac{4NF_d}{a_d}, \\ L_d &= 6d^3 a_d^{-4} B_d^{1/4}, & H_d &= d(\sqrt{d}+1)L_d, & C_d &= 2d(2dH_d)^{1/d}. \end{aligned}$$

Then, for every such K ,

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq C_d \delta^{1/(6d)}} \quad (1.1)$$

In particular, $C_d \leq d^{15}$, so the following simple numerical version holds in every dimension $d \geq 3$:

$$\boxed{D(K, \mathcal{E}_d) - 1 \leq d^{15} \left(\frac{1}{2} - a(K) \right)^{1/(6d)}} \quad (1.1a)$$

Equivalently, $\delta \leq (\varepsilon/d^{15})^{6d}$ guarantees a Banach–Mazur factor at most $1 + \varepsilon$. All constants are numerical functions of dimension. They are deliberately conservative. A more informative small-deficit bound is

$$D(K, \mathcal{E}_d) \leq \frac{1}{1 - (2dH_d \delta^{1/6})^{1/d}}, \quad (1.2)$$

whenever

$$0 < \delta \leq \delta_d^* := \left(\frac{2^{-d}}{2dH_d} \right)^6. \quad (1.3)$$

At zero deficit the v4 equality classification applies, or the zero-defect versions of the lemmas below give the same conclusion directly.

Theorem B. A dimensional obstruction. For $d \geq 3$ and $0 < t \leq 1$, set

$$K_t = [-1, 1]^d \cap \{x : |x_1 + \cdots + x_d| \leq d - t\}, \quad c_d = \frac{1}{2} - 2^{1-d}.$$

Then

$$1 + \frac{t}{5d^2} \leq D(K_t, \mathcal{E}_d) \leq \frac{d}{d-t}, \quad (1.4)$$

and

$$\frac{1}{2} - a(K_t) = c_d \frac{2^d - 2t^{d-1}/(d-1)!}{d2^d - 2t^d/(d-1)!} \frac{2t^d}{(d-1)!2^d + 2(d-1)t^{d-1}} \sim \frac{c_d}{d!2^{d-1}} t^d. \quad (1.5)$$

Consequently no universal fixed-dimensional power exponent greater than $1/d$ is possible. No positive exponent independent of dimension can work, even when its constants are allowed to depend on dimension. The optimal exponent remains between $1/(6d)$ and $1/d$; neither theorem identifies it.

The proof combines three effective estimates: a cone-specific contact-slab bound, a quantitative matching of basis coordinates, and a mixed-volume containment argument. The hypothesis and equality class differ from those of the lower-end simplex stability problem.

2. Cone law, contact identity, and the exact upper-end defect

Sections 2–5 assume $d \geq 3$; dimensions one and two were settled in the statement.

The v3 cone law is the pushforward of $h_K(u)dS_K(u)/(d|K|)$ by $u \mapsto u/h_K(u)$. Denote it by ν_K . For independent samples define

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|, \quad B = \mathbb{E}\left|\det\begin{pmatrix} X_1 & \cdots & X_N \\ 1 & \cdots & 1 \end{pmatrix}\right|.$$

These determinant expectations are classical zonoid volumes. In the normalization $Z(\nu) = \mathbb{E}[0, X]$ and $\hat{Z}(\nu) = \mathbb{E}[0, (1, X)]$, Koshevoy–Mosler, *Lift zonoids, random convex hulls and the variability of random vectors*, Bernoulli 4 (1998), 377–399, Section 4, gives

$$|Z(\nu)| = A/d!, \quad |\hat{Z}(\nu)| = B/(d+1)!.$$

Consequently the cone-law identity used here reads

$$a(K) = |\hat{Z}(\nu_K)|/|Z(\nu_K)|.$$

The determinant representation is established background; the restriction to cone laws and the endpoint inequality, equality analysis and stability are separate assertions. See the [primary author PDF](#), printed p. 387, Theorem 4.1 and equation (4.3).

We import the v3 identity $a(K) = B/(NA)$, affine invariance, and the balanced Rademacher inequality. These hold for arbitrary convex bodies, by the surface-area formulation and the continuity argument proved there. The law is even for symmetric K .

Put K in symmetric John position:

$$B_2^d \subseteq K \subseteq \sqrt{d}B_2^d. \quad (2.1)$$

Then ν_K is supported in the unit ball and $|K| \geq \kappa_d$. The law can also be sampled by a regular boundary contact $Y \in \partial K$, with density

$$\frac{\langle Y, n_K(Y) \rangle}{d|K|} d\mathcal{H}^{d-1}(Y), \quad X = \frac{n_K(Y)}{\langle Y, n_K(Y) \rangle}.$$

The nonregular boundary has surface measure zero. The divergence theorem gives

$$\langle Y, X \rangle = 1, \quad \mathbb{E}[YX^\top] = I/d. \quad (2.2)$$

For every unit vector v , since $|Y| \leq \sqrt{d}$,

$$\mathbb{E}|\langle v, X \rangle| \geq \frac{1}{d\sqrt{d}}.$$

Conditioning on successive samples and using distance to their span, this implies

$$A = \mathbb{E} \|X_1 \wedge \cdots \wedge X_d\| \geq d^{-3d/2} \geq a_d. \quad (2.3)$$

Indeed, for every fixed proper subspace choose a unit vector perpendicular to it; the conditional expected distance of the next sample to that subspace is at least $d^{-3/2}$. Multiplying these lower bounds proves (2.3), with dependent partial tuples contributing zero harmlessly. Hadamard gives $A \leq 1$.

For an N -tuple, let c_i be its signed horizontal cofactors and put $M = \sum |c_i|$. Their relation is $\sum c_i X_i = 0$. Since the points lie on ∂K° , the Minkowski norm with unit ball K° gives $|c_i| \leq M/2$. On $M > 0$, sort the normalized magnitudes as $t_1 \geq \cdots \geq t_N$, and put

$$\mathcal{D}(t) = \frac{1}{2} - \mathbb{E}_\epsilon \left| \sum_i \epsilon_i t_i \right|, \quad g = \frac{1}{2} - t_1.$$

Define expressions involving t to be zero on $M = 0$. Sign averaging the lifted determinant, using evenness, gives the exact identity

$$\Delta := \mathbb{E}[M\mathcal{D}(t)] = NA\delta, \quad \mathbb{E}M = NA \leq N. \quad (2.4)$$

This includes singular tuples. Let $\phi = Mt_4$ be the fourth-largest absolute cofactor, set to zero if all cofactors vanish.

3. Robust balanced equality and the cone-specific half-mass gate

Lemma 3.1. For balanced normalized coefficients as above,

$$\mathcal{D}(t) \geq \min\{t_4/2, g/4\}. \quad (3.1)$$

Their ℓ_1 -distance to the entire coefficient equality set (at most three nonzero entries, or one entry equal to one half) is at most $8(N-3)\mathcal{D}(t)$.

Proof. On the capped simplex $\{s_i \geq 0, \sum s_i = 1, s_i \leq 1/2\}$, the convex function $\mathbb{E}_\epsilon |\sum \epsilon_i s_i|$ is at most one half by the v3 inequality. Let $y = (1/4, 1/4, 1/4, 1/4, 0, \dots)$, whose value is $3/8$. Set $\alpha = \min(4t_4, 2g)$. If positive, $\alpha < 1$, and $(t - \alpha y)/(1 - \alpha)$ remains in the capped simplex: nonnegativity uses $\alpha \leq 4t_4$; for the first four entries the upper cap is equivalent to $t_i + \alpha/4 \leq 1/2$; for the others it follows from $\alpha \leq 2g$. Convexity yields a defect at least $\alpha/8$, proving (3.1). If $\alpha = 0$, (3.1) is immediate. To reach the half-mass branch, increase t_1 by g and subtract that mass from the other entries, costing $2g$. To reach three-point support, remove the tail of mass $q \leq (N-3)t_4$ and redistribute it in the first three entries without exceeding the cap, costing $2q$. Their available capacity is $1/2 + q$. Use the first construction when the minimum in (3.1) is $g/4$, and the second otherwise. This proves the distance bound. $\square$

The half-mass branch cannot be dropped: $(1/2, 1/6, 1/6, 1/6)$ has zero defect and four positive entries. Exact exposedness in v4 removes it at equality, but does not supply a quantitative margin. We instead use contact geometry.

Lemma 3.2. Corner-slab surface area. If $Q \subseteq [-1, 1]^d$ is any convex body, the regular boundary part where some three coordinates are within $r > 0$ of independently chosen signed cube faces has area at most

$$d2^{d+1} \binom{d}{3} r^2. \quad (3.2)$$

Proof. Fix a triple and its signs and write $\ell = \min(r, 2)$. Each coordinate projection of the relevant boundary part is contained in a box of $(d-1)$ -volume at most $2^{d-3}\ell^2$. If the deleted coordinate is outside the triple, the sharper bound is $2^{d-4}\ell^3$, which is no larger. For any measurable boundary part H , the area formula and convexity give

$$\int_H |n_i| d\mathcal{H}^{d-1} \leq 2|\pi_i H|.$$

Almost every fiber has at most an upper and a lower boundary point with nonzero normal component. Parallel boundary segments have zero such component. Since $\sum_i |n_i| \geq 1$, summing gives area at most $d2^{d-2}r^2$ for one signed triple. There are $8\binom{d}{3}$ choices, proving (3.2), also when $r \geq 2$. $\square$

Lemma 3.3. Weighted exclusion of long near-half-mass circuits. For $0 < \eta \leq 1/4$ and $\rho > 0$,

$$\mathbb{E}[M \mathbf{1}_{\{t_4 > \rho, g \leq \eta\}}] \leq Z_d(\eta/\rho)^2. \quad (3.3)$$

The event is understood on $M > 0$, as already implied by $t_4 > \rho$.

Proof. Take a union over the distinguished index k with $|c_k|/M \geq 1/2 - \eta$, and condition on the other d samples, arranged as columns of a matrix U . If $\det U = 0$, this event at k is impossible. Otherwise put $D_0 = |\det U|$, $X_k = U\alpha$, and $S = \sum |\alpha_j|$. Cofactor identities give

$$|c_k| = D_0, \quad |c_j| = D_0|\alpha_j|, \quad M = D_0(1 + S), \quad \frac{|c_k|}{M} = \frac{1}{1 + S}.$$

On this event $M \leq 4D_0$ and $S - 1 \leq 8\eta$. The condition $t_4 > \rho$ supplies at least three other coordinates with $|\alpha_j| > \rho$.

Set $Q = U^\top K$. Because the columns of U belong to K° , $Q \subseteq [-1, 1]^d$ and $|Q| = D_0|K|$. Under a primal linear map T , normalized cone mass is preserved and the scaled normal becomes $T^{-\top}X$; hence $\nu_Q = (U^{-1})_*\nu_K$. This follows directly from the regular-boundary formulas

$$n' = \frac{T^{-\top}n}{|T^{-\top}n|}, \quad dH' = |\det T| |T^{-\top}n| dH, \quad h' = h/|T^{-\top}n|.$$

For every regular contact $z \in \partial Q$ generating the conditional sample α , $\langle \alpha, z \rangle = 1$, so

$$\sum_j |\alpha_j| (1 - \operatorname{sgn}(\alpha_j) z_j) = S - 1 \leq 8\eta.$$

The three substantial coordinates therefore put z within $r = 8\eta/\rho$ of three signed cube faces. Lemma 3.2 bounds their surface area. The boundary cone density on Q is $\langle z, n_Q(z) \rangle / (d|Q|) \leq \sqrt{d}/(dD_0|K|)$, so the conditional probability is at most

$$\frac{2^{d+1}\sqrt{d}\binom{d}{3}}{D_0|K|} (8\eta/\rho)^2.$$

Multiplication by $M \leq 4D_0$ cancels the determinant exactly. Integrate the conditioning variables, sum over N indices, and use $|K| \geq \kappa_d$. The resulting constant is Z_d . $\square$

This cancellation treats arbitrarily ill-conditioned conditional matrices. The estimate uses regular primal contacts, not strong exposedness of polar points, so atoms caused by flat faces and nonatomic cone laws are both covered.

Lemma 3.4. Effective short-cofactor estimate.

$$\boxed{\mathbb{E}\phi \leq F_d \delta^{2/3}}. \quad (3.4)$$

Proof. First suppose $0 < \delta < 1/4$. For $\rho \in [\delta, 1/4]$, put $\eta = \delta^{1/3}\rho^{2/3} \leq \rho$. In the weighted tail $T(\rho) = \mathbb{E}[M\mathbf{1}_{\{t_4 > \rho\}}]$, the part with $g > \eta$ costs at most $4\Delta/\eta$ by Lemma 3.1. The other part costs at most $Z_d(\eta/\rho)^2$ by Lemma 3.3. Using (2.4),

$$T(\rho) \leq (4N + Z_d)\delta^{2/3}\rho^{-2/3}.$$

Since $t_4 \leq 1/4$, Tonelli gives $\mathbb{E}\phi = \int_0^{1/4} T(\rho) d\rho$. The interval below δ costs at most $N\delta \leq N\delta^{2/3}$; the remaining integral of $\rho^{-2/3}$ is at most 3. This gives (3.4). If $\delta \geq 1/4$, use $\mathbb{E}\phi \leq N/4 \leq N\delta^{2/3}$. If $\delta = 0$, for fixed $\rho > 0$ let $\eta \downarrow 0$ in the same split, with $\eta \leq \min(1/4, 2\rho)$, to get $T(\rho) = 0$. Tonelli proves (3.4) at zero without compactness. $\square$

4. A quantitative matching of line and plane blocks

The exact rank-two decomposition underlying this lemma has a matroid antecedent: van der Pol–Walsh–Wigal, [arXiv:2502.03673v2](#), Lemma 5.2, classify $U_{3,4}$ -minor-free matroids as direct sums of rank-at-most-two matroids. The estimate below concerns small expected cofactor magnitudes and supplies a conditioned basis and quantitative approximation to such blocks.

Lemma 4.1. Let ν be any Borel probability law supported in the unit ball, with $A \geq a$, $0 < a \leq 1$, and suppose its expected fourth absolute cofactor is at most ε . Put $q = 4(d+1)\varepsilon/a$. If $0 < q \leq 1$, there are a basis matrix U and a partition of its coordinate indices into singletons and pairs, with coordinate block union $\mathcal{W}$, such that

$$\begin{aligned} |\det U| &\geq a/2, \quad \|U\| \leq \sqrt{d}, \quad \|U^{-1}\| \leq 2\sqrt{d}/a, \\ \mathbb{E} \operatorname{dist}(U^{-1}X, \mathcal{W}) &\leq 6d^3a^{-4}q^{1/4}. \end{aligned} \tag{4.1}$$

For $\varepsilon = 0$, the same conclusion holds with zero distance. All statements include nonatomic laws and singular sampled tuples.

Proof. For a deterministic candidate basis $b = (b_1, \dots, b_d)$, define its section cost

$$Q(b) = \int \phi(b, x) d\nu(x) + \sum_{j=1}^d \iint \phi(b_1, \dots, \hat{b}_j, \dots, b_d, x, y) d\nu(x) d\nu(y).$$

Tonelli gives $\mathbb{E}_b Q(b) \leq (d+1)\varepsilon$. Since $|\det b| \leq 1$, the event $|\det b| \geq a/2$ has probability at least $a/2$. Markov gives $\mathbb{P}(Q > q) \leq a/4$. Thus choose $U = (b_i)$ with $D := |\det U| \geq a/2$ and $Q(U) \leq q$. The norm bounds follow from unit-length columns and inverse rows, each of length at most $1/D$. Cramer's rule also gives $|\alpha_i| \leq 1/D$ for $\alpha = U^{-1}X$.

Let s_3 be the third-largest coordinate magnitude. The absolute cofactors of $(b_1, \dots, b_d, X)$ are D times $(1, |\alpha_1|, \dots, |\alpha_d|)$. If $s_3 \leq 1$, its fourth largest is at least Ds_3 ; otherwise it is at least D , and $s_3 \leq 1/D$. In both cases

$$\mathbb{E}s_3 \leq q/D^2. \tag{4.2}$$

Fix $0 < t \leq 1$ and set $r = Dt^2/2$. Break coordinate ties by index. For each pair ij , let E_{ij} be the event that its coordinates are the largest two, both have magnitude at least t , and $s_3 \leq r$. Write $p_{ij} = \nu(E_{ij})$.

For distinct i, j, k , take $x \in E_{ij}$, $y \in E_{jk}$, with coordinates α, β . In the tuple consisting of the basis except b_j , followed by x, y , the cofactors attached to x, y, b_i, b_k have, up to signs, the values

$$D\beta_j, \quad D\alpha_j, \quad D(\alpha_i\beta_j - \alpha_j\beta_i), \quad D(\alpha_k\beta_j - \alpha_j\beta_k).$$

These are determinant identities without a rank assumption. The first two have magnitude at least Dt . Each of the last two has magnitude at least $D(t^2 - r/D) = Dt^2/2$, since the unwanted coordinate is at most r and the other factor at most $1/D$. Hence

$$p_{ij}p_{jk} \leq \frac{2q}{Dt^2}.$$

Pairs with mass exceeding $w = \sqrt{2q/D}/t$ form a matching. Take its pairs as plane blocks and every remaining index as a line block. Discarded pairs have total mass at most $\binom{d}{2}w$.

Outside the discarded-pair events and $\{s_3 > r\}$, a point is within $\sqrt{d}t$ of its block: if its second coordinate is below t , use the block containing its largest coordinate; otherwise use its matched pair, whose remaining coordinates are at most $r \leq t$. Every point has coordinate norm at most $\sqrt{d}/D$. By (4.2) and Markov,

$$\mathbb{E} \text{dist}(\alpha, \mathcal{W}) \leq \sqrt{d}t + \frac{\sqrt{d}}{D} \left[\binom{d}{2} \frac{\sqrt{2q/D}}{t} + \frac{2q}{D^3 t^2} \right]. \quad (4.3)$$

Take $t = q^{1/4}$. Since $q \leq 1$ and $D \geq a/2$, the right side is at most

$$\sqrt{d} [1 + 4 \binom{d}{2} a^{-3/2} + 32a^{-4}] q^{1/4} \leq 6d^3 a^{-4} q^{1/4}$$

for $d \geq 3$. This proves (4.1).

At $\varepsilon = 0$, choose $Q(U) = 0$ on the determinant-good event. Equation (4.2) makes every sample two-sparse almost surely. If two positive-mass two-coordinate classes shared an index, the displayed omitted-basis tuple would have four nonzero cofactors on their product, contradicting its zero section cost. Thus the positive-mass pairs form a matching and the coordinate distance is zero. This also proves the zero case. $\square$

The basis is selected with simultaneous control of all required sections. Its condition number is bounded explicitly. No limiting choice of a basis or unique product decomposition is assumed.

5. From approximate normals to an actual product body

Lemma 5.1. Suppose Q is a full-dimensional centered symmetric convex body, $Q \subseteq RB_2^d$, and $\mathbb{R}^d = \bigoplus_j W_j$ is an orthogonal sum. Set $\mathcal{W} = \bigcup_j W_j$ and $P = \prod_j \text{proj}_{W_j} Q$, with m blocks. If $s = \mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W})$, then

$$|P|/|Q| \leq (1 + R(\sqrt{m} + 1)s)^d. \quad (5.1)$$

Putting $v = R(\sqrt{m} + 1)s$, if $2dv < 1$,

$$(1 - (2dv)^{1/d})P \subseteq Q \subseteq P. \quad (5.2)$$

The conversion from relative volume loss to containment is the standard nested-body cap argument; compare Böröczky–Henk, *Cone-volume measure and stability*, Lemma 3.1(i), printed p. 8. Central symmetry permits the constant 2 in place of the centered-body constant e in that result.

Proof. The projection product contains Q , is symmetric, and lies in $\sqrt{m}RB_2^d$. Its support function agrees with h_Q on every block. For $x \in \partial Q^\circ$, choose a closest block projection $z \in \mathcal{W}$. Support functions are globally Lipschitz with their circumradii, so

$$0 \leq h_P(x) - 1 = h_P(x) - h_Q(x) \leq R(\sqrt{m} + 1)|x - z|.$$

The cone-law definition, homogeneity, and the first mixed-volume formula give

$$\frac{V(Q[d-1], P)}{|Q|} - 1 = \mathbb{E}_{\nu_Q}[h_P(X) - 1] \leq v.$$

Minkowski's first inequality gives (5.1). Consequently $\theta = 1 - |Q|/|P| \leq 1 - (1+v)^{-d} \leq dv$.

For completeness, a symmetric nested-body cap converts this relative volume loss to containment. Let λ be maximal with $\lambda P \subseteq Q$; then $0 < \lambda \leq 1$. If $\lambda < 1$, a touching supporting direction u and point $p \in P$ satisfy $h_Q(u) = \lambda h_P(u)$, $\langle u, p \rangle = h_P(u)$. The image

$$\lambda p + (1-\lambda)(P \cap \{z : \langle u, z \rangle \geq 0\})$$

lies in P by convexity and outside $\text{int } Q$ by support. Symmetry makes its volume $(1-\lambda)^d |P|/2$. Hence $\theta \geq (1-\lambda)^d/2$, proving (5.2). No inverse cone-volume uniqueness theorem is used. $\square$

Proof of Theorem A. Use (2.1). Lemma 3.4 gives $\varepsilon = F_d \delta^{2/3}$. By (2.3), apply Lemma 4.1 with $a = a_d$, so $q = B_d \delta^{2/3}$. For $\delta \leq \delta_d^*$, $q \leq 1$: indeed $H_d \geq B_d^{1/4}$, so $B_d(2^{-d}/(2dH_d))^4 \leq 1$. The resulting coordinate law satisfies

$$\mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) \leq L_d \delta^{1/6}.$$

Apply the primal map $U^\top$. Its body $Q = U^\top K$ lies in dB_2^d , because $\|U\| \leq \sqrt{d}$, and its cone law is the coordinate law just obtained. Its blocks are orthogonal coordinate lines and planes. Lemma 5.1 gives $v \leq H_d \delta^{1/6}$, with a product $P \in \mathcal{E}_d$, proving (1.2). In the range (1.3), the quantity $r = (2dH_d \delta^{1/6})^{1/d}$ is at most $1/2$, so

$$D(K, \mathcal{E}_d) - 1 \leq \frac{r}{1-r} \leq 2(2dH_d)^{1/d} \delta^{1/(6d)}.$$

For larger deficits, a universal elementary bound is $D(K, \mathcal{E}_d) \leq d$. In John position, $d^{-1/2}[-1, 1]^d \subseteq K \subseteq d^{1/2}[-1, 1]^d$, and the cube belongs to $\mathcal{E}_d$. If $\delta \geq \delta_d^*$, the constant in (1.1) satisfies

$$C_d \delta^{1/(6d)} \geq 2d(2dH_d)^{1/d} (\delta_d^*)^{1/(6d)} = d,$$

which covers $D - 1 \leq d - 1$. At zero use the zero cases of the lemmas or v4. For $d = 1, 2$, every symmetric body is itself an allowed factor. This completes the proof. $\square$

For the stated simpler constant, the cube inscribed in the unit ball gives $\kappa_d \geq (2/\sqrt{d})^d$. Using $N \leq 2d$ and $\binom{d}{3} \leq d^3$, one obtains

$$\begin{aligned} Z_d &\leq 2^{10} d^{d/2+9/2}, & F_d &\leq 2^{12} d^{d/2+9/2}, & B_d &\leq 2^{15} d^{(5/2)d+11/2}, \\ H_d &\leq 12 \cdot 2^{15/4} d^{(69/8)d+47/8}, & C_d^d &\leq 2^d (24 \cdot 2^{15/4}) d^{(77/8)d+55/8}. \end{aligned}$$

Since $2^d \leq d^d$ and $24 \cdot 2^{15/4} < 512 \leq d^6$ for $d \geq 3$,

$$C_d^d \leq d^{(85/8)d+103/8} \leq d^{15d},$$

where the last inequality is $35d \geq 103$. This proves (1.1a).

6. Proof of the dimensional obstruction

We give the full-class argument, since comparing only to the cube would not establish a distance lower bound to arbitrary planar-factor products.

Let $D \in \mathcal{E}_d$ and suppose $K_t \subseteq D \subseteq (1+s)K_t$. Any defining Banach–Mazur sandwich can be rescaled into this form, preserving $\mathcal{E}_d$. Write

$$D = F_1 + \cdots + F_m, \quad \mathbb{R}^d = W_1 \oplus \cdots \oplus W_m, \quad F_j \subset W_j, \quad \dim W_j \in \{1, 2\}.$$

The factors are centered symmetric and full-dimensional in their subspaces. Let P_j be the projection onto W_j along the remaining summands, and set $a_{ji} = h_{F_j}(e_i) \geq 0$. Then

$$P_j^2 = P_j, \quad \text{tr } P_j = \dim W_j, \quad \sum_j P_j = I, \quad 1 \leq \sum_j a_{ji} \leq 1+s. \quad (6.1)$$

Every cube vertex $\sigma \in \{\pm 1\}^d$ except the two all-equal ones belongs to K_t . Since $P_j \sigma \in F_j$, the nonnegative quantities $a_{ji} - \sigma_i(P_j \sigma)_i$ sum to at most s . Therefore

$$|(P_j \sigma)_i - a_{ji} \sigma_i| \leq s. \quad (6.2)$$

For any coordinate k , assign both signs among the other coordinates; this is possible since $d \geq 3$. Both vertices obtained by changing only σ_k are retained. Subtract (6.2) for them to obtain

$$|(P_j - \text{diag}(a_{j1}, \dots, a_{jd}))_{ik}| \leq s. \quad (6.3)$$

This controls all entries even when the original decomposition is badly conditioned.

Suppose $s \leq 1/(4d)$, and write $p_{ji} = (P_j)_{ii}$. Idempotence gives $|p_{ji}(p_{ji} - 1)| \leq (d-1)s^2$. Thus some $z_{ji} \in \{0, 1\}$ satisfies

$$|p_{ji} - z_{ji}| \leq 2(d-1)s^2.$$

Inside $[0, 1]$, this follows from $q(1-q) \geq q/2$ for distance $q \leq 1/2$ to an endpoint; outside the interval it follows directly. The integer trace and sum identities in (6.1), with errors less than one half, give $\sum_i z_{ji} = \dim W_j$ and $\sum_j z_{ji} = 1$. Hence $J_j = \{i : z_{ji} = 1\}$ partitions the indices into sets of size one or two. For $i \notin J_j$, (6.3) gives $a_{ji} \leq s + 2(d-1)s^2$.

Let $u = (1, \dots, 1)$. Choose a retained vertex with plus signs on J_j and a minus sign outside it. Equation (6.2) implies

$$h_{F_j}(u) \geq \sum_i a_{ji} - 3ds - 4d(d-1)s^2 \geq \sum_i a_{ji} - 4ds.$$

Summing gives $h_D(u) \geq d - 4d^2s$, while containment in $(1+s)K_t$ gives $h_D(u) \leq d - t + ds$. Therefore $t \leq 5d^2s$. If $s > 1/(4d)$, the same lower bound follows from $t \leq 1$. This proves the lower part of (1.4) for every product decomposition. The upper part follows from $(1-t/d)[-1, 1]^d \subseteq K_t \subseteq [-1, 1]^d$.

For the exact deficit, the two removed corner simplices each have volume $t^d/d!$, and every coordinate facet has area $\alpha = 2^{d-1} - t^{d-1}/(d-1)!$. The two new area-normal vectors are $\pm t^{d-1}u/(d-1)!$, with area times support number $\beta = (d-t)t^{d-1}/(d-1)!$. Put

$$V_t = 2^d - 2t^d/d!, \quad D_t = d2^d - 2t^d/(d-1)!, \quad p = \frac{2^d - 2t^{d-1}/(d-1)!}{D_t}, \quad r = \frac{2(d-t)t^{d-1}}{(d-1)!D_t}.$$

The cone law has direction pairs $\pm e_i$, each with total probability p , and $\pm u/(d-t)$, with total probability r . Within pairs signs are uniform; $dp + r = 1$. Direct type enumeration gives

$$A = d!p^{d-1} \left(p + \frac{dr}{d-t} \right).$$

Only N -tuples containing all $d + 1$ different direction types have a nonzero Rademacher defect. Tuples with fewer than d types have zero cofactors; those with exactly d types have a repeated parallel direction and a two-term cofactor relation. For the all-distinct tuple, cofactor magnitudes are 1 and d copies of $q = 1/(d - t)$. Averaging the sign multiplying 1 gives $\mathbb{E} \max(1, |q \sum_{i=1}^d \epsilon_i|)$. For $t \leq 1$, only the two all-equal sign patterns exceed 1, so its value is $1 + 2^{1-d}(dq - 1)$. The defect is $c_d t/(d - t)$. The probability of all-distinct types is $N! p^d r$, giving

$$\delta_t = \frac{c_d p r t}{(d - t)p + d r}.$$

Substitute p, r to get (1.5). Also $p \leq 1/d$, hence $\delta_t \leq c_d t^d/(d! 2^{d-1})$. Combining this with (1.4) excludes every exponent greater than $1/d$ as $t \downarrow 0$. In dimension three,

$$\delta_t = \frac{t^3(8 - t^2)}{8(24 - t^3)(4 + t^2)}.$$

This proves Theorem B. $\square$

7. Comparison with primary cone-volume and stability literature

The class $\mathcal{E}_d$ is the familiar class of affine Cartesian products of centrally symmetric intervals and planar bodies, called Weil bodies in Saroglou, [On the shape of a convex body with respect to its second projection body](#), pp. 2–4. That paper relates the class to Weil’s polytopal classification for homothety under the second projection-body operator and gives a three-dimensional zonoid endpoint with symmetric-cylinder equality (Theorem 1.2). These use a different functional from $a(K)$; the common extremal class alone does not make their statements equivalent to entry005 v4’s equality theorem or Theorem A here.

The relevant antecedent is Böröczky–Lutwak–Yang–Zhang, [The logarithmic Minkowski problem](#), JAMS 26 (2013), 831–852. Its subspace concentration condition and complementary-subspace clause characterize existence of symmetric cone-volume measures. They do not supply the near-short-circuit estimate (3.4), a quantitative matching into dimensions at most two, or the present invariant’s endpoint.

Böröczky–Henk, [Cone-volume measure and stability](#), Adv. Math. 306 (2017), 24–50, Theorem 1.1 gives complementary-factor stability with containment exponent $1/(5d)$ from almost saturated cone mass on an **exact** proper subspace; its constants and smallness threshold are stated as dimension-dependent, rather than numerical formulas. Its U-functional stability targets parallelotopes. Our equality class includes all symmetric planar factors, and our hypothesis is determinant magnitude, not the U-functional’s independence indicator. Thus that theorem is not an equivalent endpoint result.

The exact-subspace premise cannot be inferred here: for even integers $p \rightarrow \infty$, smooth strictly convex ℓ_p balls converge to the cube and have $a \rightarrow 1/2$, yet their cone-volume measures give zero mass to every proper linear subspace. On each open coordinate octant the Gauss map is a diffeomorphism with positive curvature; the omitted coordinate-zero parts have surface measure zero, proving absolute continuity. Our method instead produces an expected distance to blocks and recovers an actual product by Lemma 5.1.

Böröczky–De, [Stable solution of the logarithmic Minkowski problem in the case of hyperplane symmetries](#), J. Differential Equations 298 (2021), 298–322, obtains quantitative inverse cone-volume estimates under Coxeter symmetry and a tube concentration margin. The reducible-group margin degenerates at concentration on a proper group-invariant product block, and the symmetry hypothesis does not cover arbitrary symmetric planar factors. Their anisotropic-squeeze example also explains

why normalization is necessary. Abdallah–Mérigot, [On the reconstruction of convex sets from random normal measurements](#), gives a surface-area inverse estimate of order $1/d$ with area and rotundity control. Cone-volume data additionally contains the unknown support function; Lemma 5.1 avoids solving that inverse problem.

Böröczky–De also discuss volume-normalized unconditional bodies obtained by truncating vertices of a cube ([published version](#), p. 302). This provides related geometric context. Theorem B independently computes the upper-end scalar deficit $1/2 - a(K_t)$ and proves a distance lower bound against every affine product of symmetric one- and two-dimensional bodies. A comparison with one fixed cube would not establish that full-class lower bound.

Saroglou’s [Volumes of projection bodies of some classes of convex bodies](#) is a primary projection-body comparison. Only its publication record and abstract were available; a full-text comparison remains incomplete. The recent Liu–Xiong–Yang mixed-volume stability paper, DOI [10.1016/j.aim.2026.110886](#), is relevant to broader mixed-volume stability questions. No theorem from either of these two papers is imported as a dependency. The present literature comparison is confined to the primary statements described above.

8. Dependencies, reproducibility, and open questions

The affine calculus and cone-law representation are inherited from entry 005 versions 2–3; the exact equality class is the version 4 theorem. The pinned sources are listed below. The quantitative contact-slab, matching, and product recovery arguments in Sections 3–5 are supplied in full here. In particular, the lower-end simplex modulus and the balanced-recursion spectral theorem are separate results and are not inputs to this proof.

The accompanying exact programs test 4,930 balanced rational coefficient multisets, 750 omitted-basis cofactor identities, and 16 corner-truncation facet-minor calculations. They use only integers and rational arithmetic, and explicit exceptions keep the checks active under Python optimization. These finite regressions supplement the written analytic proof. The package records source hashes and separates regenerated outputs from shipped reference reports.

The constants are deliberately coarse. The optimal universal exponent remains between $1/(6d)$ and $1/d$ for $d \geq 3$. Sharpening the matching estimate and the conversion of normal concentration to product containment may reduce the gap.

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Versions 2–4 credit the projection-volume product and simplex computations inherited from OpenAI family 088, pinned at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. This supplement uses the inherited identities through the explicitly cited entry 005 sources.

III Weighted rank two matching

A sharp square-root matching theorem for bounded laws

This note strengthens the matching lemma in the completed upper-end stability supplement. It is separate research; it does not alter that supplement.

Statement and conventions

Let $d \geq 3$, let ν be a Borel probability measure supported in the Euclidean unit ball, and let $X_1, \dots, X_{d+1}$ be independent samples. Write

$$A = \mathbb{E}|\det(X_1, \dots, X_d)|.$$

For a tuple of $d+1$ vectors, let ϕ denote the fourth-largest of the $d+1$ absolute d -dimensional cofactors. Suppose

$$A \geq a > 0, \quad \mathbb{E}\phi \leq \varepsilon, \quad q = 4(d+1)\varepsilon/a.$$

Hadamard's inequality implies $a \leq 1$.

Theorem 1 (square-root matching). There are a basis matrix U whose columns lie in the support of ν , and a partition of its coordinates into singletons and pairs, such that the union $\mathcal{W}$ of the resulting coordinate blocks satisfies

$$|\det U| \geq a/2, \quad \|U\| \leq \sqrt{d}, \quad \|U^{-1}\| \leq 2\sqrt{d}/a,$$

and, if $0 \leq q \leq 1$,

$$\mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) \leq 5d^2a^{-2}\sqrt{q}. \quad (1)$$

The conclusion includes nonatomic laws and singular sampled tuples. The power $1/2$ cannot be increased under these hypotheses, even for even finite laws and fixed dimension.

Proof of Theorem 1

For a candidate sampled basis $b = (b_1, \dots, b_d)$, put

$$Q(b) = \int \phi(b_1, \dots, b_d, x) d\nu(x) + \sum_{j=1}^d \iint \phi(b_1, \dots, \hat{b}_j, \dots, b_d, x, y) d\nu(x) d\nu(y).$$

Tonelli's theorem gives $\mathbb{E}_b Q(b) \leq (d+1)\varepsilon$. Since $D(b) = |\det b| \leq 1$ and its expectation is at least a , the event $D(b) \geq a/2$ has probability at least $a/2$. If $\varepsilon > 0$, Markov's inequality gives $\mathbb{P}(Q(b) > q) \leq a/4$. These two events therefore intersect. If $\varepsilon = 0$, $Q(b) = 0$ almost everywhere and the determinant-good event again intersects it. Fix such a basis $U = (b_i)$, and write

$$D = |\det U| \geq a/2, \quad R = D^{-1}.$$

Each inverse row has length at most D^{-1} , by its cofactor-wedge formula and the unit-length bound on the other columns. Consequently $\|U^{-1}\| \leq \sqrt{d}/D$, $\|U\| \leq \sqrt{d}$, and the coordinates $\alpha = U^{-1}X$ obey $\max_i |\alpha_i| \leq R$.

Let $s_3(\alpha)$ be the third-largest coordinate magnitude. The absolute cofactors of $(b_1, \dots, b_d, X)$ are D times $(1, |\alpha_1|, \dots, |\alpha_d|)$. If $s_3 \leq 1$, the fourth largest is at least Ds_3 ; otherwise it is at least D , and $s_3 \leq D^{-1}$. Thus

$$\mathbb{E}s_3 \leq Q_0/D^2 \leq q/D^2, \quad (2)$$

where Q_0 is the one-sample term of $Q(U)$.

For every α , retain its two largest coordinates, breaking ties by index, and set all others to zero. Denote the resulting vector by $\tilde{\alpha}$. This is a Borel map, it does not enlarge the coordinate bounds, and

$$\|\alpha - \tilde{\alpha}\|_\infty \leq s_3(\alpha), \quad |\alpha - \tilde{\alpha}| \leq \sqrt{d} s_3(\alpha). \quad (3)$$

We first control the two-sample sections after this truncation. For independent coordinate vectors α, β , the absolute cofactors of the tuple omitting b_j and appending $U\alpha, U\beta$ are the absolute values of

$$D\alpha_j, \quad D\beta_j, \quad D(\alpha_i\beta_j - \alpha_j\beta_i) \quad (i \neq j), \quad (4)$$

up to signs and reordering. This is the polynomial determinant identity obtained by applying U^{-1} to every column.

Replacing both vectors by their truncations changes either linear entry by at most $D(s_3(\alpha) + s_3(\beta))$. For every bilinear entry, telescoping the two products and using all coordinate bounds $R = D^{-1}$ gives a change at most

$$2DR(s_3(\alpha) + s_3(\beta)) = 2(s_3(\alpha) + s_3(\beta)).$$

Taking absolute values and the fourth-largest entry are both 1-Lipschitz for the maximum norm on these finite lists. Therefore the truncated section expectation $\tilde{Q}_j$ satisfies

$$\tilde{Q}_j \leq Q_j + 4\mathbb{E}s_3 \leq q + 4q/D^2 =: \tilde{q}. \quad (5)$$

No regularity or atom assumption is used here.

For each unordered pair ij , let E_{ij} be the Borel event that i, j are the selected two largest coordinates. Define its *weighted* mass by

$$m_{ij} = \mathbb{E} \left[\min(|\alpha_i|, |\alpha_j|) \mathbf{1}_{E_{ij}} \right]. \quad (6)$$

Zero second coordinates contribute zero, so arbitrary tie choices there are harmless. Consider distinct i, j, k . On $E_{ij} \times E_{jk}$, the two truncated vectors are exactly supported on ij and jk . Put $r_\alpha = \min(|\alpha_i|, |\alpha_j|)$ and $r_\beta = \min(|\beta_j|, |\beta_k|)$. Four entries in (4) have absolute values

$$D|\alpha_j|, \quad D|\beta_j|, \quad D|\alpha_i\beta_j|, \quad D|\alpha_j\beta_k|.$$

Since $r_\alpha, r_\beta \leq R$ and $D \leq 1$, each of these is at least $D^2 r_\alpha r_\beta$. Integrating the fourth cofactor over this product event gives

$$D^2 m_{ij} m_{jk} \leq \tilde{Q}_j \leq \tilde{q}. \quad (7)$$

In particular, edges of weight strictly greater than $w = \sqrt{\tilde{q}}/D$ cannot share a vertex. They form a matching. Use those edges as pair blocks and every remaining coordinate as a singleton block. Every discarded edge has weight at most w .

If the selected top pair of α is a matching edge, its truncated vector lies in its plane block. Otherwise, its distance to the block containing its largest coordinate is at most the smaller retained coordinate. By (3) and the triangle inequality,

$$\text{dist}(\alpha, \mathcal{W}) \leq \sqrt{d} s_3(\alpha) + \min(|\alpha_i|, |\alpha_j|) \mathbf{1}_{\{ij \text{ discarded}\}}.$$

Hence (2), (5), and (6) give

$$\mathbb{E} \text{dist}(\alpha, \mathcal{W}) \leq \frac{\sqrt{d} q}{D^2} + \binom{d}{2} \frac{\sqrt{q + 4q/D^2}}{D} \leq \frac{\sqrt{d} + \sqrt{5} \binom{d}{2}}{D^2} \sqrt{q}. \quad (8)$$

The last step uses $q \leq 1$ and $D \leq 1$. Finally, $D \geq a/2$, $\sqrt{5} \leq 5/2$, and $\sqrt{d} \leq d$ imply

$$4 \left(\sqrt{d} + \sqrt{5} \binom{d}{2} \right) \leq 4d + 5d(d-1) \leq 5d^2,$$

which proves (1). At $q = 0$, (7) says all positive-weight edges form a matching, (2) gives exact two-sparsity almost everywhere, and (8) has zero right-hand side. This proves the zero case directly.

A sharp obstruction under the intermediate hypotheses

Lemma 2 (cofactor vanishing for a block union). Let $\mathbb{R}^d = \bigoplus_j W_j$, where $\dim W_j \in \{1, 2\}$. For every $d+1$ vectors in $\bigcup_j W_j$, the fourth absolute cofactor is zero.

Proof. If the tuple has rank below d , all cofactors vanish. Otherwise assign each vector to one block containing it, assigning a zero vector arbitrarily. If block j receives n_j vectors, full rank requires $n_j \geq \dim W_j$ for every block. As their sum is $d+1$, exactly one block receives one extra vector and the others receive precisely their dimensions. The unique relation on the tuple is supported on that oversampled block, which receives at most three vectors. Cofactors are its relation coefficients, so at most three cofactors are nonzero. $\square$

Lemma 3 (a quantitative distance witness). If $z_1, \dots, z_{d+1}$ are in the unit ball, then, for every block union $\mathcal{V}$ in Lemma 2,

$$\sum_{i=1}^{d+1} \text{dist}(z_i, \mathcal{V}) \geq \phi(z_1, \dots, z_{d+1}). \quad (9)$$

Proof. Project each z_i orthogonally onto a closest block and call the resulting vector y_i . Then $|y_i| \leq 1$ and $|z_i - y_i| = \text{dist}(z_i, \mathcal{V})$. For each cofactor, multilinear telescoping and Hadamard's inequality bound its change by the sum of these distances over its d columns, hence by the full sum. The fourth absolute cofactor is 1-Lipschitz in the list of cofactors, and it vanishes for the y_i by Lemma 2. $\square$

Now put

$$v_{12} = (e_1 + e_2)/\sqrt{2}, \quad v_{23} = (e_2 + e_3)/\sqrt{2}, \quad 0 < p \leq 1/(d+2), \quad w = (1 - 2p)/d.$$

Let ν_p give total mass w to each antipodal pair $\{\pm e_i\}$, and mass p to each of $\{\pm v_{12}\}, \{\pm v_{23}\}$, distributing each antipodal mass equally. This is an even law in the unit ball, and $w \geq p$. Sampling the d axis directions in any order gives

$$A(\nu_p) \geq d!w^d \geq a_d := d!/(d+2)^d > 0. \quad (10)$$

Any repeated direction gives a relation supported on its two repeated columns. If the tuple has rank below d , all cofactors vanish; otherwise its unique relation is that repeated-direction relation, so at most two cofactors are nonzero. Thus such a tuple has fourth cofactor zero. Among distinct-direction $d+1$ -tuples, the only set with positive fourth cofactor is

$$(e_1, e_3, e_4, \dots, e_d, v_{12}, v_{23}). \quad (11)$$

Indeed, omitting an outside axis gives rank below d ; omitting one of the two new directions leaves a three-term circuit; and omitting e_1 or e_3 likewise leaves the three-term circuit on e_2, e_3, v_{23} or e_1, e_2, v_{12} . In (11), the four nonzero absolute cofactors are $1/2, 1/2, 1/\sqrt{2}, 1/\sqrt{2}$, and all other cofactors are zero. Its fourth cofactor is $1/2$. Therefore, exactly,

$$\varepsilon_p := \mathbb{E}_{\nu_p^{d+1}} \phi = \frac{(d+1)!}{2} w^{d-1} p^2. \quad (12)$$

Signs and permutations do not change this value.

Applying Lemma 3 to (11), and using that every direction has mass at least p , gives for *every* direct sum of line and plane blocks

$$\mathbb{E}_{\nu_p} \text{dist}(X, \mathcal{V}) \geq p/2. \quad (13)$$

This lower bound is independent of the choice, angles, or uniqueness of the decomposition. For any basis U with unit-length columns, let $\mathcal{V} = U\mathcal{W}$. Since $\|U\| \leq \sqrt{d}$,

$$\mathbb{E} \text{dist}(U^{-1}X, \mathcal{W}) \geq \frac{1}{\sqrt{d}} \mathbb{E} \text{dist}(X, \mathcal{V}) \geq \frac{p}{2\sqrt{d}}. \quad (14)$$

As $p \downarrow 0$, (12) is a positive constant depending on d times $p^2 + O_d(p^3)$, while (10) is uniformly positive. Thus an estimate with power $\gamma > 1/2$ and constants depending only on a, d is impossible under the hypotheses of Theorem 1.

The obstruction is about arbitrary bounded even laws. No claim is made that these particular laws are cone laws of centrally symmetric bodies. The completed cone-specific half-mass argument must still be retained when applying the matching theorem to convex bodies.

Consequence for the existing proof chain

Keeping the completed estimate $\mathbb{E}\phi \leq F_d \delta^{2/3}$, its determinant lower bound $a = a_d^{\text{John}}$, and its geometry lemma unchanged, (1) gives

$$s \leq 5d^2 (a_d^{\text{John}})^{-2} \sqrt{4(d+1)F_d/a_d^{\text{John}}} \delta^{1/3}.$$

The same cap-to-containment step therefore yields a proved upper-end power $1/(3d)$, with explicit constants obtained by this substitution. The root assembly is responsible for stating and simplifying the final global constant. The intermediate square-root matching step is sharp, so improving it further requires additional cone-law information or a different geometric route.

Verification scope

`check_matching_improvement.py` checks the cofactor formula, the truncation perturbation inequality, the adjacent-edge lower bound, and the exact finite-law obstruction formula using rational arithmetic. For the last check it uses the rational variant $v_{ij} = (e_i + e_j)/2$, whose fourth cofactor is $1/4$ and whose expectation is $(d+1)!w^{d-1}p^2/4$, avoiding any floating-point square roots. These finite checks support the polynomial identities; the measure-theoretic selection and integral proof above do not depend on enumeration.

IV Direct geometric recovery

Direct quantitative recovery from off-block normals

Date: 7 October 2026. This is a separate research note. It does not alter the completed upper-end stability supplement.

1. Statement

Let $d \geq 2$, and let $Q \subset \mathbb{R}^d$ be an origin-symmetric ($Q = -Q$), full-dimensional convex body such that

$$rB_2^d \subset Q \subset RB_2^d, \quad 0 < r \leq R.$$

Let $\mathbb{R}^d = \bigoplus_{j=1}^m W_j$ be an orthogonal decomposition into nonzero subspaces, let $\mathcal{W} = \bigcup_j W_j$, and put

$$P = \prod_{j=1}^m \text{proj}_{W_j} Q, \quad I = \int_{\partial Q} \text{dist}(n_Q(x), \mathcal{W}) d\mathcal{H}^{d-1}(x).$$

Regular boundary points suffice in this integral; the omitted set has surface measure zero. Thus

$$I = \int_{S^{d-1}} \text{dist}(u, \mathcal{W}) dS_Q(u).$$

Write $\kappa_k = |B_2^k|$, including $\kappa_0 = 1$, and set

$$A_{d,r,R} = \frac{2}{\kappa_{d-2}} \left(\frac{2(\sqrt{2}R + r)}{r} \right)^{d-2}.$$

Theorem. The product built from the actual block projections satisfies

$$d_H(Q, P) \leq m(A_{d,r,R} I)^{1/(d-1)}, \quad (1.1)$$

$$d_{\text{BM}}(Q, P) - 1 \leq \frac{m}{r} (A_{d,r,R} I)^{1/(d-1)}. \quad (1.2)$$

Equivalently, for the cone-law measurement

$$s = \mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W}) = \frac{I}{d|Q|},$$

one has the explicit estimate

$$\boxed{d_{\text{BM}}(Q, P) - 1 \leq \frac{m}{r} \left(A_{d,r,R} d\kappa_d R^d s \right)^{1/(d-1)}}. \quad (1.3)$$

No atomicity, polyhedrality, smoothness, or uniqueness of a product decomposition is required. The dimensions of the blocks do not enter the proof. If every block has dimension one or two, P is in the equality class of the completed theorem, so its distance to the entire class has the same upper bound. For $m = 1$, $P = Q$, and all claims are immediate.

2. A planar mixed-normal lemma

Lemma. Let $C \subset \mathbb{R}^2$ be a full-dimensional compact convex set. Suppose C contains points $(-b_0, 0)$ and $(0, -a_0)$, where $a_0, b_0 > 0$, while $\text{dist}(0, C) \geq t > 0$. Then

$$\int_{\partial C} \min(|n_1|, |n_2|) d\mathcal{H}^1 \geq t. \quad (2.1)$$

Proof. Intersect with the closed negative quadrant. Choose the closest intersection points to the origin on its coordinate axes and write them as $(-b, 0)$ and $(0, -a)$, with $a, b > 0$. They exist because the original two points are in C , and positivity follows from the distance assumption. For $-b \leq x \leq 0$, let

$$f(x) = \max\{y : (x, y) \in C \cap (-\infty, 0]^2\}.$$

The segment connecting the two chosen points proves that this function is defined on the entire interval. Convexity shows that f is concave; it is continuous, $f(-b) = 0$, and $f(0) = -a$. The choice of b gives $f(x) < 0$ for $-b < x \leq 0$. Concavity and $f \leq 0$ imply that f is nonincreasing. Its open graph belongs to the boundary of the original C , because both quadrant constraints are inactive there.

Put $g = -f$. This is a nonnegative, nondecreasing convex function, with $g(-b) = 0$ and $g(0) = a$. It is absolutely continuous and its almost-everywhere derivative is nonnegative and nondecreasing. At regular graph points the unit outward normal is $(g'(x), 1)/\sqrt{1 + g'(x)^2}$, so the graph contribution is

$$\int_{-b}^0 \min(1, g'(x)) dx.$$

Since g' is nondecreasing, let $x_* \in [-b, 0]$ be a crossing point of the level one, with the natural endpoint choices if $g' \geq 1$ everywhere or $g' \leq 1$ everywhere. Integration before and after the crossing gives

$$\int_{-b}^0 \min(1, g') dx = g(x_*) - x_* = \min_{[-b, 0]} (g(x) - x).$$

Each graph point $(x, -g(x)) \in C$ is at Euclidean distance at least

t from the origin. Its distance in the ℓ_1 norm is $g(x) - x$, hence the displayed minimum is at least t . All other boundary contributions are nonnegative. This proves (2.1). Endpoint vertical segments, if present, have zero mixed-normal contribution and do not change the argument. $\square$

3. Two-factor recovery by thickening a planar section

Fix an orthogonal bipartition $\mathbb{R}^d = E \oplus F$ into nonzero subspaces and put

$$P_{E,F} = \text{proj}_E Q \times \text{proj}_F Q, \quad I_{E,F} = \int_{\partial Q} \min(|\text{proj}_E n|, |\text{proj}_F n|) d\mathcal{H}^{d-1}.$$

We claim

$$d_H(Q, P_{E,F})^{d-1} \leq A_{d,r,R} I_{E,F}. \quad (3.1)$$

Since $Q \subset P_{E,F}$, choose a point $p \in P_{E,F}$ where $t = \text{dist}(p, Q) = d_H(Q, P_{E,F})$ is attained. If $t = 0$ the claim is immediate. Choose $q_A, q_B \in Q$ with

$$\text{proj}_E q_A = \text{proj}_E p, \quad \text{proj}_F q_B = \text{proj}_F p.$$

Then $a = p - q_A \in F$ and $b = p - q_B \in E$ are nonzero and orthogonal. Define the two-dimensional plane

$$L = \text{span}\{b, a\}, \quad e_1 = b/|b| \in E, \quad e_2 = a/|a| \in F, \quad T = L^\perp.$$

Because each component of p is a projection of a point of Q , $|p| \leq \sqrt{2}R$. Also $t \leq |p| \leq \sqrt{2}R$, because $0 \in Q$. Set

$$\eta = \frac{t}{2(\sqrt{2}R + r)} \in (0, 1).$$

For every $z \in T$ with $|z| < \eta r$, the points

$$q_{A,z} = (1 - \eta)q_A + z, \quad q_{B,z} = (1 - \eta)q_B + z$$

belong to the interior of Q : each is the convex combination of a point of Q and the interior-ball point z/η . They share an affine plane parallel to L with

$$p_z = (1 - \eta)p + z.$$

Relative to p_z , the endpoints have coordinates $(0, -(1 - \eta)|a|)$ and $(-(1 - \eta)|b|, 0)$. Furthermore

$$\text{dist}(p_z, Q) \geq t - |p - p_z| \geq t - \eta\sqrt{2}R - |z| \geq t/2. \quad (3.2)$$

The planar section $C_z = Q \cap (p_z + L)$ is full-dimensional in its plane and contains both endpoints. Applying the planar lemma gives

$$\int_{\partial C_z} \min(|n_{C_z} \cdot e_1|, |n_{C_z} \cdot e_2|) d\mathcal{H}^1 \geq t/2. \quad (3.3)$$

The linear projection $\pi_T : \partial Q \rightarrow T$ has tangential $(d - 2)$ -Jacobian $|\text{proj}_L n|$. For almost every transverse parameter, its slice has outward normal

$$n_{C_z} = \frac{\text{proj}_L n}{|\text{proj}_L n|}$$

at almost every regular intersection point. The coarea formula therefore gives, with zero integrand when $\text{proj}_L n = 0$,

$$\begin{aligned}
I_{E,F} &\geq \int_{\partial Q} \min(|n \cdot e_1|, |n \cdot e_2|) d\mathcal{H}^{d-1} \\
&\geq \int_{\{|z| < \eta r\} \subset T} \int_{\partial C_z} \min(|n_{C_z} \cdot e_1|, |n_{C_z} \cdot e_2|) d\mathcal{H}^1 dz \\
&\geq \kappa_{d-2}(\eta r)^{d-2} t/2 \\
&= \frac{\kappa_{d-2}}{2} \left(\frac{r}{2(\sqrt{2}R + r)} \right)^{d-2} t^{d-1}.
\end{aligned}$$

This is (3.1). For $d = 2$, the transverse space has dimension zero, and the same formula is interpreted with $\kappa_0 = 1$.

The coarea argument applies to arbitrary convex bodies: their boundaries are countably rectifiable and regular almost everywhere. The singular boundary has zero $(d-1)$ -measure and therefore zero weighted coarea contribution. For almost every transverse parameter, the intersection of the regular boundary is the section boundary up to length-zero sets. The ball thickening ensures that all sections used here have nonempty relative interior. Flat facets and nonatomic surface-area laws are both allowed.

4. Recovering all factors

For each j , apply (3.1) to $E = W_j$, $F = W_j^\perp$, and write the resulting two-factor product as P_j . For any unit vector n ,

$$\min(|\text{proj}_{W_j} n|, |\text{proj}_{W_j^\perp} n|) \leq \text{dist}(n, \mathcal{W}).$$

Indeed, if the largest block component is the j -th one, the right-hand side is exactly the complementary norm; if another block component is largest, the right-hand side is at least the j -th component norm. Consequently

$$e_j := d_H(Q, P_j) \leq (A_{d,r,R} I)^{1/(d-1)}. \quad (4.1)$$

Fix any $p \in P$, start with $q_0 \in Q$, and successively replace the j -th block coordinate of q_{j-1} by that of p . Call the new point z_j . It belongs to P_j , because its j -th component is feasible in the projection of Q , while the complementary component is the projection of $q_{j-1} \in Q$. Choose $q_j \in Q$ such that

$$q_j = z_j + \varepsilon_j, \quad |\varepsilon_j| \leq e_j.$$

Writing $\pi_j = \text{proj}_{W_j}$, the recurrence is

$$q_j = \pi_j p + (I - \pi_j)q_{j-1} + \varepsilon_j.$$

The mutually orthogonal coordinate projections commute. After all blocks have been replaced, the original q_0 contributes zero and

$$q_m - p = \sum_{j=1}^m \left(\prod_{k=j+1}^m (I - \pi_k) \right) \varepsilon_j.$$

Every displayed product is an orthogonal projection, so $|q_m - p| \leq \sum_j e_j$. Taking the maximum over $p \in P$ proves (1.1).

Finally, if $e = d_H(Q, P)$, support functions satisfy

$$h_P(u) \leq h_Q(u) + e \leq (1 + e/r)h_Q(u)$$

for unit u , because $h_Q(u) \geq r$. Hence

$$(1 + e/r)^{-1}P \subset Q \subset P.$$

This proves (1.2), and $|Q| \leq \kappa_d R^d$ gives (1.3).

5. Consequences and limits

The weighted corner cuts in `geometry-next.md`, and their all-product distance obstruction imported from the audited supplement, show that the exponent $1/(d-1)$ in (1.3) cannot be improved in general when the blocks have dimension one or two. Their radii can be kept uniformly bounded above and below. Thus this direct recovery theorem closes the geometry-only exponent gap; it does not identify the sharp exponent of the original upper-end deficit.

The selected basis in the completed rank lemma already gives an explicit inradius sufficient to apply this theorem. If its columns have Euclidean norm at most one and determinant $D \geq a_d/2$, every row of its inverse has norm at most $1/D$ by the cofactor-wedge bound. Therefore

$$\|U^{-1}\| \leq \sqrt{d}/D, \quad U^T K \supset \frac{a_d}{2\sqrt{d}} B_2^d$$

when K is in symmetric John position. The same transformed body satisfies $U^T K \subset dB_2^d$. Thus the existing estimate $s \leq L_d \delta^{1/6}$ yields an improved geometric exponent $\delta^{1/[6(d-1)]}$, with constants obtained directly from (1.3). The direct geometric estimate and its use in the stronger matching assembly passed an independent analytic model audit. It does not change the completed release and does not rely on the extra-factor estimate refuted in `geometry-next.md`.

6. Stronger homothetic recovery without an inradius constant

The interior-ball thickening can be replaced by a thickening with Q itself. This removes the inradius from the small-error bound.

Keep $Q = -Q$, $Q \subset RB_2^d$, and the orthogonal block decomposition from above. Define

$$T = \left(2^{2d-1} dR s\right)^{1/(d-1)}, \quad \lambda = \max\{\alpha \geq 0 : \alpha P \subset Q\}.$$

Theorem. For arbitrary nonzero block dimensions,

$$\boxed{1 - \lambda \leq mT.} \tag{6.1}$$

If $mT \leq 1/2$, then

$$d_{\text{BM}}(Q, P) - 1 \leq 2mT. \tag{6.2}$$

When all block dimensions are one or two, the global whole-class estimate is

$$\boxed{D(Q, \mathcal{E}_d) - 1 \leq 2md \left(2^{2d-1} d R s \right)^{1/(d-1)}}. \quad (6.3)$$

The global estimate uses the symmetric John bound only for the regime $mT > 1/2$. It is an estimate to the entire affine product class; in that regime it does not claim control of distance to the one fixed product P .

6.1. Bipartition lemma For an orthogonal bipartition $E \oplus F$, put $P_{E,F}$ and $I_{E,F}$ as in Section 3, and set

$$\lambda_{E,F} = \max\{\alpha : \alpha P_{E,F} \subset Q\}, \quad \tau = 1 - \lambda_{E,F}.$$

We prove

$$\boxed{\tau^{d-1} \leq 2^{2d-1} R \frac{I_{E,F}}{|Q|}}. \quad (6.4)$$

Both bodies contain the origin in their interiors, and $Q \subset P_{E,F}$, so $\lambda_{E,F} \in (0, 1]$. The claim is immediate when $\tau = 0$. Otherwise compactness of the support-function ratio gives a unit vector u such that

$$h_Q(u) = \lambda_{E,F} h_{P_{E,F}}(u).$$

Write $H = h_{P_{E,F}}(u) > 0$, and choose $p \in P_{E,F}$ with $u \cdot p = H$. Choose $q_A, q_B \in Q$ sharing respectively the E and F components of p . As before,

$$a = p - q_A \in F, \quad b = p - q_B \in E, \quad L = \text{span}\{b, a\}, \quad T_0 = L^\perp.$$

The two vectors a, b are nonzero: the support gap at p is $\tau H > 0$. Furthermore $u_L = \text{proj}_L u \neq 0$, because

$$u \cdot a = H - u \cdot q_A \geq \tau H > 0.$$

Set $\eta = \tau/4 \in (0, 1/4)$. For any $z \in \text{int}_{T_0}(\text{proj}_{T_0} Q)$, choose an interior lift $q \in \text{int } Q$ with $\text{proj}_{T_0} q = z$. Such a lift exists because a surjective linear map sends the interior of a convex body onto the interior of its image. Consider

$$p_q = (1 - \eta)p + \eta q, \quad q_{A,q} = (1 - \eta)q_A + \eta q, \quad q_{B,q} = (1 - \eta)q_B + \eta q.$$

The endpoints belong to the interior of Q , share a plane parallel to L with p_q , and relative to p_q have the same negative-axis form as in Section 3. Central symmetry gives $u \cdot q \geq -h_Q(u)$. Consequently

$$\begin{aligned} u \cdot p_q - h_Q(u) &= \tau H - \eta(H - u \cdot q) \\ &\geq \tau H - \eta(H + h_Q(u)) \\ &\geq \tau H - 2\eta H = \tau H/2. \end{aligned}$$

On this fixed affine plane, only the component u_L of the normal varies. Thus the distance of p_q to the planar section is at least

$$\frac{\tau H}{2|u_L|}.$$

Apply the planar lemma and then the same weighted coarea identity as in Section 3. The transverse parameters now fill

$$(1 - \eta) \operatorname{proj}_{T_0} p + \eta \operatorname{proj}_{T_0} Q,$$

whose $(d - 2)$ -volume is $\eta^{d-2} |\operatorname{proj}_{T_0} Q|$. The boundary of this projection has measure zero. Every interior parameter has an interior lift, so its section is full-dimensional; no measurable selection of the lift is needed, since the lower bound holds for each such section. It follows that

$$I_{E,F} \geq \eta^{d-2} |\operatorname{proj}_{T_0} Q| \frac{\tau H}{2|u_L|}. \quad (6.5)$$

For any fixed transverse parameter w , its fiber in Q is contained in a planar rectangle with side lengths at most

$$\frac{2h_Q(u)}{|u_L|} \leq \frac{2H}{|u_L|} \quad \text{and} \quad 2R.$$

The first follows from $-h_Q(u) \leq u_L \cdot x_L + u_{T_0} \cdot w \leq h_Q(u)$; the second follows from $Q \subset RB_2^d$ in the perpendicular direction inside L . Fubini therefore yields

$$|Q| \leq \frac{4RH}{|u_L|} |\operatorname{proj}_{T_0} Q|. \quad (6.6)$$

Substitute (6.6) into (6.5), and use $\eta = \tau/4$:

$$\frac{I_{E,F}}{|Q|} \geq \frac{\tau^{d-1}}{84^{d-2}R} = \frac{\tau^{d-1}}{2^{2d-1}R}.$$

This proves (6.4), also in dimension two with the zero-dimensional transverse volume convention.

6.2. Combining the homothetic inclusions For each single-block bipartition $P_j = \operatorname{proj}_{W_j} Q \times \operatorname{proj}_{W_j^\perp} Q$, let $\lambda_j = \max\{\alpha : \alpha P_j \subset Q\}$. The pointwise normal inequality from Section 4 and $I = d|Q|s$ give

$$1 - \lambda_j \leq T. \quad (6.7)$$

We claim

$$\left(\prod_{j=1}^m \lambda_j \right) P \subset Q. \quad (6.8)$$

Fix $p \in P$, write $p_j = \operatorname{proj}_{W_j} p$, and prescribe the adjusted coordinates

$$\tilde{p}_j = \left(\prod_{k < j} \lambda_k \right) p_j.$$

Each adjusted coordinate lies in its factor projection, since that factor is convex and contains zero. Start with $q_0 = 0 \in Q$. For $j = 1, \dots, m$, replace the j -th block of q_{j-1} by $\tilde{p}_j$ to obtain $z_j \in P_j$, and then set $q_j = \lambda_j z_j \in Q$. The j -th block of q_m is

$$\left(\prod_{k=j}^m \lambda_k \right) \tilde{p}_j = \left(\prod_{k=1}^m \lambda_k \right) p_j.$$

Therefore $q_m = (\prod_j \lambda_j) p \in Q$, proving (6.8). Since the λ_j lie in $(0, 1]$,

$$1 - \prod_j \lambda_j \leq \sum_j (1 - \lambda_j) \leq mT.$$

This proves (6.1). In the small-error regime it gives

$$(1 - mT)P \subset Q \subset P, \quad d_{\text{BM}}(Q, P) - 1 \leq \frac{mT}{1 - mT} \leq 2mT,$$

which is (6.2). For the whole-class estimate, the same bound applies when $mT \leq 1/2$, and otherwise symmetric John's theorem gives $D(Q, \mathcal{E}_d) - 1 \leq d - 1 \leq 2mdT$. This proves (6.3).

6.3. Original endpoint implication The completed proof's final coordinate body satisfies $Q \subset dB_2^d$. Thus (6.2) or (6.3) can be used without paying for the small determinant of the selected basis. In particular, the existing $s \leq L_d \delta^{1/6}$ improves the deficit exponent from $1/(6d)$ to $1/[6(d-1)]$. Any independent improvement in the matching-stage power can be combined with this proof, after checking its precise inputs.

The $1/(d-1)$ power in (6.3) is sharp in fixed dimension by the corner cut bodies and the all-product distance lower bound in geometry-next.md. The new proof does not infer a sharp original-deficit exponent from that intermediate sharpness.

The direct recovery proof uses thickening by the body itself and a shared support direction to cancel the conditioning loss. Its small-error estimate needs no inradius constant and has geometric power $1/(d-1)$. The optimal original-deficit exponent remains unresolved.

V Half mass obstruction for actual cone laws

Sharpness of the cone-specific half-mass gate

7 October 2026. This note concerns the short-cofactor gate in Sections 2–3 of the completed upper-end stability manuscript. It does not modify that manuscript or assert an improved stability exponent.

Result

For every fixed dimension $d \geq 3$, the estimate

$$\mathbb{E}\phi \leq F_d \delta^{2/3}$$

has the best possible power under its stated hypotheses, even among centrally symmetric polytopes already in symmetric John position. Here $\delta = 1/2 - a(K)$, and ϕ is the fourth-largest absolute horizontal cofactor of $d + 1$ independent cone-law samples. Neither the corner-slab area power two nor the weighted near-half-mass tail power two can be improved without adding hypotheses or changing the intermediate quantity.

The obstruction is an actual cone law. It does not use a freely prescribed even law on a polar boundary.

1. A product of a truncated three-cube with intervals

For $0 < t \leq 1$, define

$$H_t = [-1, 1]^3 \cap \{|x_1 + x_2 + x_3| \leq 3 - t\}, \quad K_{d,t} = H_t \times [-1, 1]^{d-3}.$$

The interpretation in dimension three is $K_{3,t} = H_t$. The unit ball is contained in $K_{d,t}$: its first-three-coordinate sum has absolute value at most $\sqrt{3} < 2 \leq 3 - t$. The body is contained in the cube and hence in $\sqrt{d}B_2^d$. Its coordinate-facet contacts $\pm e_i$, with weights $1/2$ for each sign, give the symmetric John identity $\sum_i (e_i e_i^T) = I$. Thus the unit ball is its John ellipsoid; there is no hidden change of normalization in the cofactor calculation.

Put

$$Z = 24 - t^3, \quad p = \frac{8 - t^2}{Z}, \quad r = \frac{(3 - t)t^2}{Z}, \quad q = \frac{1}{3 - t}.$$

The cone law of H_t has three direction pairs $\pm e_i$, each with total probability p , and one pair $\pm q(e_1 + e_2 + e_3)$, with total probability r . Signs within each pair are uniform. In particular $3p + r = 1$. Indeed, $|H_t| = 8 - t^3/3$. Each coordinate facet has area $4 - t^2/2$ and support height one, giving pair cone mass $8 - t^2$. Each new triangular facet has area $\sqrt{3}t^2/2$, support height $(3 - t)/\sqrt{3}$, and scaled normal $\pm q(e_1 + e_2 + e_3)$. Its two-sign pair cone mass is $(3 - t)t^2$. The normalization is $3|H_t| = Z$.

The product cone law is supported on the same four direction pairs in the first three coordinates, together with $\pm e_4, \dots, \pm e_d$. Its pair probabilities are

$$\begin{aligned} P &= \frac{3}{d}p \quad \text{for each of the first three axes,} \\ R &= \frac{3}{d}r \quad \text{for the diagonal pair,} \\ w &= \frac{1}{d} \quad \text{for every remaining axis.} \end{aligned}$$

To verify the dimension factors directly, a facet lifted from the three-dimensional factor has its area-times-height and total body volume both multiplied by 2^{d-3} ; division by $d|K_{d,t}|$ instead of $3|H_t|$

multiplies the corresponding normalized mass by $3/d$. Every interval-axis pair has area-times-height $|K_{d,t}|$, and therefore normalized mass $1/d$. This argument uses ordinary facet cone-volume mass and does not assume that an arbitrary discrete measure is realizable.

2. Exact fourth-cofactor expectation

Call the unsigned direction pairs types. The $d + 1$ types span dimension d , and their unique linear dependence is supported on the first three axes and the diagonal. A $d + 1$ -tuple with fewer than d distinct types has all horizontal cofactors zero. A tuple with exactly d distinct types has one repeated parallel direction. If its rank is d , its kernel is generated by the relation between those two columns and has just two nonzero cofactors; if its rank is smaller, all cofactors vanish.

Consequently $\phi > 0$ occurs exactly when all $d + 1$ types occur. On that event the absolute cofactors are

$$1, q, q, q, \underbrace{0, \dots, 0}_{d-3},$$

regardless of the independent signs. Since $q \leq 1/2$, their fourth-largest member is q . The event probability is

$$\Pi = (d + 1)! P^3 R w^{d-3} = \frac{81(d + 1)!}{d^{d+1}} p^3 r.$$

Therefore

$$\mathbb{E}\phi = \frac{81(d + 1)!}{d^{d+1}} \frac{(8 - t^2)^3 t^2}{(24 - t^3)^4} \sim \frac{(d + 1)!}{8d^{d+1}} t^2. \quad (2.1)$$

3. Exact deficit

For completeness this deficit can be computed without invoking the dimension-weighted product identity. The determinant expectation is

$$A = d! w^{d-3} P^2 (P + 3Rq) = \frac{27d!}{d^d} p^2 (p + 3rq).$$

The first term uses all d axis types; the three remaining nonzero terms replace one of the first three axes by the diagonal. Replacing any other axis by the diagonal gives zero determinant.

Tuples with fewer than d types have zero cofactor mass. Tuples with exactly d types have zero balanced Rademacher defect, because their two nonzero cofactor magnitudes agree. On the all-distinct event, sign averaging gives

$$\mathbb{E}_\epsilon |\epsilon_0 + q(\epsilon_1 + \epsilon_2 + \epsilon_3)| = 1 + \frac{1}{4}(3q - 1).$$

Here only the two all-equal patterns in the last three signs have $|q(\epsilon_1 + \epsilon_2 + \epsilon_3)| > 1$; all other patterns have value $q < 1$. Its unnormalized cofactor defect is consequently

$$\frac{1 + 3q}{2} - \left(1 + \frac{1}{4}(3q - 1)\right) = \frac{3q - 1}{4} = \frac{t}{4(3 - t)}.$$

Using $\Delta = (d + 1)A\delta$, this proves

$$\delta_{d,t} = \frac{3}{d} \frac{t^3(8 - t^2)}{8(24 - t^3)(4 + t^2)} \sim \frac{1}{32d} t^3. \quad (3.1)$$

In particular the normalization agrees with the completed manuscript's dimension-three formula. Combining (2.1) and (3.1), for every $\beta > 2/3$,

$$\frac{\mathbb{E}\phi}{\delta_{d,t}^\beta} \sim \frac{(d+1)!}{8d^{d+1}} (32d)^\beta t^{2-3\beta} \longrightarrow \infty.$$

No dimension-dependent finite constant can repair an exponent $\beta > 2/3$ in this intermediate estimate, in any fixed $d \geq 3$.

4. Sharpness of the weighted tail and contact-slab powers

On the all-distinct event,

$$M = 1 + 3q = \frac{6-t}{3-t}, \quad t_4 = \frac{1}{6-t}, \quad g = \frac{1}{2} - \frac{1}{M} = \frac{t}{2(6-t)}.$$

All other tuples have $t_4 = 0$. Choose $\rho = 1/7$ and $\eta_t = t/[2(6-t)]$. Then $t_4 > \rho$ on every all-distinct tuple and $\eta_t \leq 1/10 < 1/4$. Thus

$$\mathbb{E}[M \mathbf{1}_{\{t_4 > \rho, g \leq \eta_t\}}] = M\Pi = \frac{81(d+1)!}{d^{d+1}} \frac{(8-t^2)^3 t^2 (6-t)}{(24-t^3)^4} \sim \frac{3(d+1)!}{4d^{d+1}} t^2.$$

Since $t = 12\eta_t/(1+2\eta_t)$,

$$\lim_{t \downarrow 0} \frac{\mathbb{E}[M \mathbf{1}_{\{t_4 > 1/7, g \leq \eta_t\}}]}{\eta_t^2} = \frac{108(d+1)!}{d^{d+1}} > 0. \quad (4.1)$$

Therefore the exponent two in the weighted near-half-mass bound cannot be increased, even with ρ fixed and even for these actual cone laws.

The corner-slab area estimate is independently sharp. The two diagonal facets of $K_{d,t}$ lie in the respective three-coordinate corner slabs of width t : if $x_1 + x_2 + x_3 = 3-t$, then each $x_i \geq 1-t$, with the analogous negative assertion. Their total regular boundary area is

$$\sqrt{3} 2^{d-3} t^2.$$

Any area bound with a power strictly greater than two would fail as $t \downarrow 0$. Flat facets, rather than exceptional nonregular contacts, already realize this obstruction. The completed projection-area and determinant-cancellation argument therefore cannot acquire a better power merely by a stronger universal patch-area estimate.

5. A different geometric quantity does avoid this loss on the family

The preceding obstruction concerns an intermediate moment. It does not exclude a stronger final stability exponent through another argument. Let $C_d = [-1, 1]^d$ and consider the actual cone support surplus

$$v = \mathbb{E}_{\nu_{K_{d,t}}} [h_{C_d}(X) - 1].$$

Axis samples have zero surplus. At a diagonal sample the surplus is $3q - 1 = t/(3-t)$, and its probability is R . Consequently

$$\boxed{v = \frac{3t^3}{d(24-t^3)}, \quad \frac{v}{\delta_{d,t}} = \frac{8(4+t^2)}{8-t^2} \leq \frac{40}{7}.} \quad (5.1)$$

The actual mixed-volume error used in product recovery is therefore linear in δ on this sharp half-mass family. Bounding the support surplus by a support-function Lipschitz constant times distance to the coordinate line/plane union discards a factor of order t here: the diagonal samples stay a positive distance from that union, while their support surplus tends to zero. A proof of a global joint block-selection and support-surplus estimate would be a different route; this note proves no such estimate.

One loss-free weighted quantity is already available for arbitrary balanced coefficients. Since $t_4 \leq 1/4$ and $g \leq 1/2$,

$$t_4 g \leq \min(t_4/2, g/4) \leq \mathcal{D}(t).$$

Hence the exact deficit identity gives

$$\boxed{\mathbb{E}[\phi g] \leq \Delta = (d+1)A\delta.} \tag{5.2}$$

On the truncation family this weighted moment has order t^3 , rather than the order t^2 of $\mathbb{E}\phi$. Equation (5.2) is not by itself a geometric stability theorem: the missing step is to select one global line/plane decomposition while retaining an actual support-surplus weight instead of unweighted sparsity distance.

Verification scope

The analytic argument above proves the all-dimensional statements. An optional exact-rational regression in `check_halfmass_sharpness.py` checks the formulas by direct determinant/type and sign enumeration in dimensions three and four. It does not replace the proof and imposes no polytope restriction on the completed upper stability theorem.

VI Weighted cuts and geometric obstruction

Geometry gate: exact obstructions and one restricted improvement

Date: 7 October 2026. This note leaves the completed quantitative theorem unchanged. It isolates what can and cannot be inferred from the inputs of its product-recovery lemma. Every assertion below is proved explicitly; the general recovery theorem subsequently obtained is in the separate note `geometry-rigidity.md`.

1. Measurements and a cancellation worth preserving

Let Q be a centered full-dimensional convex body, let $\mathbb{R}^d = \bigoplus_j W_j$ be an orthogonal decomposition, and write $\mathcal{W} = \bigcup_j W_j$, $P = \prod_j \text{proj}_{W_j} Q$. Set

$$s(Q, \mathcal{W}) = \mathbb{E}_{\nu_Q} \text{dist}(X, \mathcal{W}), \quad v(Q, P) = \mathbb{E}_{\nu_Q} [h_P(X) - 1].$$

Because distance to the union of linear subspaces is homogeneous, the cone weight cancels exactly:

$$s(Q, \mathcal{W}) = \frac{1}{d|Q|} \int_{S^{d-1}} \text{dist}(u, \mathcal{W}) dS_Q(u). \quad (\text{G1})$$

There is no support-height factor left in (G1). This is valid for arbitrary surface-area measures, including atoms. The mixed-volume identity likewise is

$$v(Q, P) = \frac{1}{d|Q|} \int (h_P(u) - h_Q(u)) dS_Q(u) = \frac{V(Q[d-1], P)}{|Q|} - 1. \quad (\text{G2})$$

The completed proof bounds $v \leq R(\sqrt{m} + 1)s$, then converts relative volume loss into a containment loss. The example below distinguishes those two steps.

2. A complete exact calculation for weighted corner cuts

Fix $d \geq 3$, positive coefficients $a_i \leq 1$ with $\max_i a_i = 1$, and $0 < t \leq \min_i a_i$. Let

$$S = \sum_i a_i, \quad H = S - t, \quad A_* = \prod_i a_i, \quad Q_{a,t} = [-1, 1]^d \cap \{x : |\langle a, x \rangle| \leq H\}. \quad (\text{G3})$$

Both removed corner simplices are contained in their respective cube corner regions: writing $y_i = 1 - x_i$, the positive cap is $y_i \geq 0$, $\sum_i a_i y_i < t$, and its intercepts are $t/a_i \leq 1$. Their interiors are disjoint. In particular

$$V := |Q_{a,t}| = 2^d - \frac{2t^d}{d!A_*}. \quad (\text{G4})$$

For any coordinate block of size at most two, its projection is the entire coordinate cube. Indeed, at any vertex of that block choose signs among the remaining coordinates so that the full vertex is neither of the two all-equal vertices. Such a full vertex has $|\langle a, x \rangle| \leq S - 2 \min_i a_i \leq S - t$. All coordinate-block cube vertices therefore lift to points of $Q_{a,t}$, and convexity proves the claim. Hence $P = [-1, 1]^d$ for every such coordinate partition. (The same statement holds for a single coordinate projection without needing a partition.)

Each coordinate facet has area

$$\alpha_i = 2^{d-1} - \frac{t^{d-1}}{(d-1)! \prod_{k \neq i} a_k}.$$

The two new facet area-normal vectors are

$$\pm ca, \quad c = \frac{t^{d-1}}{(d-1)! A_*}. \quad (\text{G5})$$

For example, projecting a new facet onto $e_i^\perp$ gives a $(d-1)$ -simplex of volume $t^{d-1}/((d-1)! \prod_{k \neq i} a_k) = ca_i$, proving (G5) by the projection area formula. All old facet normals belong to $\mathcal{W}$, and the new cone-law sample is $\pm a/H$. Consequently the exact formulas are

$$\boxed{s(Q_{a,t}, \mathcal{W}) = \frac{2t^{d-1} \text{dist}(a, \mathcal{W})}{d(d-1)! A_* V}, \quad v(Q_{a,t}, P) = \frac{2t^d}{d(d-1)! A_* V}.} \quad (\text{G6})$$

Their ratio is particularly informative:

$$\frac{v}{s} = \frac{t}{\text{dist}(a, \mathcal{W})}. \quad (\text{G7})$$

These formulas account for every facet, rather than just the new faces. The normalization can also be checked directly:

$$2 \sum_i \alpha_i + 2cH = dV.$$

The maximal centered homothetic containment is exactly

$$\lambda P \subseteq Q_{a,t}, \quad \lambda = \frac{S-t}{S}, \quad \tau := 1 - \lambda = \frac{t}{S}. \quad (\text{G8})$$

Indeed, the sole new slab constraint forces and suffices for this factor.

3. Two sharp upper restrictions, with different exponents

For $a = (1, \dots, 1)$, write $K_t = Q_{a,t}$. If the coordinate partition has largest block dimension $b \in \{1, 2\}$, then

$$s(K_t, \mathcal{W}) = \frac{2\sqrt{d-b}}{d(d-1)! |K_t|} t^{d-1}, \quad v(K_t, P) = \frac{2}{d(d-1)! |K_t|} t^d. \quad (\text{G9})$$

Here $P = [-1, 1]^d$, $\tau = t/d$, and $\theta = 1 - |K_t|/|P| = 2t^d/(d!2^d)$. Thus the relative-volume-to-homothetic-containment exponent $1/d$ is optimal already in this nested symmetric family. It cannot be increased even when both bodies have uniformly bounded radii and a fixed positive inradius.

For the stronger conclusion involving distance to the **entire** product class, the completed and independently audited Theorem B proves

$$D(K_t, \mathcal{E}_d) - 1 \geq \frac{t}{5d^2}. \quad (\text{G10})$$

Combining (G9) and (G10) proves the following precise intermediate obstruction. No estimate

$$D(Q, \mathcal{E}_d) - 1 \leq C_{d,r,R} s(Q, \mathcal{W})^\beta$$

can hold for all bodies and all coordinate line/plane decompositions with $rB \subseteq Q \subseteq RB$ when $\beta > 1/(d-1)$. One may take $r = 2/3$, $R = \sqrt{d}$, and the same fixed partition throughout: K_t contains $(1-t/d)[-1, 1]^d$, so these bounds hold for $0 < t \leq 1$. This obstruction does **not** prove that the present geometric $1/d$ exponent is sharp as a function of s . The exact normal measurement has a different scale from relative missing volume.

4. A rigorous $1/(d-1)$ recovery theorem for this full weighted family

Assume in addition $t \leq 1/2$. Choose a coordinate block attaining the maximum projected norm of a . At least one index k is outside that block because every block has dimension at most two and $d \geq 3$. Since all coefficients are at most one,

$$A_* \leq a_k \leq \text{dist}(a, \mathcal{W}).$$

Using $V \leq 2^d$ in (G6) gives

$$s \geq \frac{t^{d-1}}{d!2^{d-1}}.$$

Since $S \geq 1$, (G8) yields

$$D(Q_{a,t}, \mathcal{E}_d) - 1 \leq \frac{t}{S-t} \leq 2t \leq \boxed{2(d!2^{d-1}s)^{1/(d-1)}}. \quad (\text{G11})$$

This restricted theorem is uniform over arbitrarily small coefficients

a_i , subject only to $t \leq \min_i a_i$: nearly coordinate normals do not destroy the improved exponent for weighted corner cuts. Together with (G9)–(G10), it identifies $1/(d-1)$ as the optimal geometry exponent within this family. It does not silently assert that the general body is a polytope or belongs to this family.

5. An obstruction to an attractive extra-factor argument

It is tempting to replace $v \leq Cs$ by $v \leq C\tau s$, with $\tau = 1 - \lambda$. That stronger intermediate assertion is false, even with fixed radius and inradius bounds and in the explicit family above.

Take $a = (1, \varepsilon, \dots, \varepsilon)$, $t = \varepsilon$, $0 < \varepsilon \leq 1/2$, and all coordinate lines as blocks. Then

$$\begin{aligned} S &= 1 + (d-1)\varepsilon, & V &= 2^d - \frac{2\varepsilon}{d!}, & \tau &= \frac{\varepsilon}{1 + (d-1)\varepsilon}, \\ s &= \frac{2\sqrt{d-1}\varepsilon}{d(d-1)!V}, & v &= \frac{2\varepsilon}{d(d-1)!V}, & \frac{v}{\tau s} &= \frac{1 + (d-1)\varepsilon}{\sqrt{d-1}\varepsilon} \rightarrow \infty. \end{aligned} \quad (\text{G12})$$

In particular no constant depending only on d , r , and R can justify this factor. Here $Q \subseteq \sqrt{d}B$ and, by (G8), $Q \supseteq (1 - \varepsilon)[-1, 1]^d \supseteq \frac{1}{2}B$.

Even the pointwise proposed estimate

$$h_P(X) - 1 \leq C\tau \operatorname{dist}(X, \mathcal{W})$$

fails at both new facet samples: the left-hand side is ε/H , whereas the distance is $\sqrt{d-1}\varepsilon/H$. The ratio before inserting τ is fixed. The small facets responsible for a missing corner can instead be macroscopic nearly coordinate facets, with a normal angle of order ε . This is an actual convex-body counterexample, not a synthetic law unrelated to its body.

6. What this resolves and what remains

The cap-to-distance step is sharp **under its volume-only input**. The separate note `geometry-rigidity.md` now proves the general $1/(d-1)$ recovery theorem under fixed radius and inradius bounds, with explicit constants. Its direct proof thickens a two-dimensional convex section and applies the coarea formula to the off-block surface integral (G1). Thus (G9)–(G10) identify the sharp geometric exponent under these stronger inputs. No extra τ factor is inserted in (G2), and the volume-only cap conversion is bypassed.

The half-mass and matching stages may also be coupled directly to the support surplus v , instead of first bounding s . Equations (G7) and (G12) show why such a coupling must preserve information about the normal direction and cannot depend only on global containment error.

The all-product distance lower bound (G10) is imported only from the completed audited theorem; its proof handles arbitrary oblique planar factor products and does not assume uniqueness of the decomposition.

7. Exact finite verification

The adjacent script `check_geometry_next.py` uses rational arithmetic to verify facet normalization, the exact v identity, squared s identities, and the near-coordinate counterexample for finitely many dimensions and rational weights. It is supplementary verification, not a proof of any universal statement. It needs only Python's standard library, uses explicit exceptions, and retains checks under `python -O`.

The corner-cut examples distinguish missing volume of order t^d from off-block normal mass of order t^{d-1} . The geometric $1/(d-1)$ estimate is established separately; the extra containment factor proposed for mixed-volume recovery is contradicted by the displayed family.

VII Slanted actual cone law obstruction

Coupled upper-end stability: a cone-law obstruction to a linear volume gate

This note is separate from the completed upper-end stability supplement. It contains proved gate obstructions and one explicitly unproved route. No claim of an improved global exponent is made here. The separate matching and geometry notes improve the matching and containment gates, respectively.

1. Precise obstruction under the selected-basis intermediate hypotheses

Let $0 < e \leq 1/2$ and put

$$Q_e = [-1, 1]^3 \cap \{|x_1 + ex_2| \leq 1, |x_2 + ex_3| \leq 1\}. \quad (1.1)$$

The body is centrally symmetric and full-dimensional, with $(1-e)[-1, 1]^3 \subset Q_e \subset [-1, 1]^3$. Its cone law has five direction pairs

$$\pm z_1 = \pm e_1, \quad \pm z_2 = \pm e_2, \quad \pm z_3 = \pm e_3, \quad \pm z_4 = \pm(1, e, 0), \quad \pm z_5 = \pm(0, 1, e). \quad (1.2)$$

Write p_i for the probability of the entire pair, with uniform signs inside each pair. Define

$$\begin{aligned} V_e &= 8 - 4e + e^2 - e^3/3, \\ C_1 = C_4 &= 2 - e/2, \quad C_2 = 2 - e, \\ C_3 &= 4 - 3e + e^2 - e^3/2, \quad C_5 = 2 - e + e^2/2, \\ p_i &= 2C_i/(3V_e). \end{aligned} \quad (1.3)$$

In particular $\sum_{i=1,4,5} p_i > 1/8$, and

$$(p_1, p_2, p_3, p_4, p_5) \longrightarrow (1/6, 1/6, 1/3, 1/6, 1/6). \quad (1.4)$$

Let ϕ be the fourth-largest absolute cofactor of a four-sample tuple, and let $A = E|\det(X_1, X_2, X_3)|$. Exact enumeration gives

$$\mathbb{E}\phi = 24p_1p_3p_4p_5e^2 \sim e^2/27, \quad (1.5)$$

$$\begin{aligned} A &= 6[p_1p_2p_3 + ep_1p_2p_5 + ep_1p_3p_4 + p_1p_3p_5 \\ &\quad + e^2p_1p_4p_5 + p_2p_3p_4 + ep_2p_4p_5 + p_3p_4p_5] \longrightarrow 2/9, \end{aligned} \quad (1.6)$$

and the scalar upper-end deficit is

$$\delta_e = \frac{1}{2} - a(Q_e) = \frac{3p_1p_3p_4p_5e^2}{A} \sim e^2/48. \quad (1.7)$$

For the basis $U=I$, the simultaneous section cost used in the matching lemma is exactly

$$Q(U) = \int \phi(e_1, e_2, e_3, x) d\nu(x) + \sum_{j=1}^3 \iint \phi(e_1, \dots, \widehat{e}_j, \dots, e_3, x, y) d\nu(x) d\nu(y) = 2p_4 p_5 e^2 \sim e^2/18. \quad (1.8)$$

Let W be any coordinate partition into lines and planes, and let $P_W = \prod_j \text{proj}_j(W_j)$. All four possible partitions satisfy

$$\begin{aligned} v_W &:= \mathbb{E}_{\nu_{Q_e}}[h_{P_W}(X) - 1] \\ &= \mathbb{E}_{\nu_{Q_e}} \text{dist}(X, \bigcup_j W_j) \in \{p_4 e, p_5 e, (p_4 + p_5)e\}, \\ \min_W v_W &= e \min(p_4, p_5) \geq e/8. \end{aligned} \quad (1.9)$$

Thus neither a coordinate-block mean-distance estimate nor a coordinate-block mixed-volume-surplus estimate with exponent greater than $1/2$ in $Q(U)$ can hold under these hypotheses. In particular, the tempting replacement

$$\min_W v_W \leq C \mathbb{E} \phi \quad \text{or} \quad \min_W v_W \leq C \delta_e \quad (1.10)$$

is false for this uniformly nondegenerate family. These particular support calculations concern the prescribed basis and its coordinate partitions. Section 3 gives separate proofs for arbitrary block choices and for distance to the entire affine equality class; neither follows merely from (1.9).

The unit-ball hypothesis of the matching lemma is obtained without changing the obstruction. Put $c = \sqrt{1+e^2}$, replace Q_e by $c Q_e$, and choose the basis with columns e_i/c . Every cone-law sample then has norm at most one, the basis determinant is $c^{-3} = (5/4)^{-3/2}$, its coordinate law remains exactly (1.2), and its transformed primal body remains Q_e . The section cost and $\mathbb{E} \phi$ are simply divided by c^3 . Its determinant expectation remains bounded below by a positive constant. Consequently the example meets the bounded-law, determinant, simultaneous-section, and controlled-basis hypotheses with constants independent of e .

2. Proof of the formulas

Fix $u = x_2$. The allowed x_1 interval has length $2-e|u|$. The allowed x_3 interval has length 2 if $|u| \leq 1-e$, and length $1+(1-|u|)/e$ otherwise. Therefore

$$|Q_e| = 2 \int_0^1 (2 - eu) \begin{cases} 2, & u \leq 1 - e, \\ 1 + (1 - u)/e, & u > 1 - e \end{cases} du = V_e.$$

On the positive $x_1=1$ facet one has $-1 \leq u \leq 0$, giving area $2-e/2$. On the positive $x_2=1$ facet the allowed x_1 and x_3 lengths are $2-e$ and 1, respectively. On the positive $x_3=1$ facet one integrates $2-e|u|$ over $-1 \leq u \leq 1-e$, giving C_3 . These coordinate facets have support number one.

The positive new facet $x_1+e x_2=1$ is parametrized by $x_1=1-eu$, $0 \leq u \leq 1$, and its allowed x_3 interval. Its area-normal vector is $C_4(1, e, 0)$; its area times support number is $C_4=2-e/2$. The other positive new facet has $x_2=1-e x_3$, $0 \leq x_3 \leq 1$, and x_1 length $2-e+e^2 x_3$. Its area-normal vector is $C_5(0, 1, e)$, with area times support number C_5 . These calculations prove the cone masses in (1.3). They also give $\sum C_i = 3V_e/2$, as required by the divergence theorem.

For four samples to have four nonzero cofactors, their four distinct direction types must be $\{1, 3, 4, 5\}$. Every other type set has a zero cofactor, and any repeated type also produces at most three nonzero cofactors. For the exceptional type set the cofactor magnitudes are

$$(1, 1, e, e^2).$$

Its fourth absolute cofactor is e^2 . Its unnormalized balanced Rademacher defect is exactly

$$\frac{2 + e + e^2}{2} - \mathbb{E}|\epsilon_1 + \epsilon_2 + e\epsilon_3 + e^2\epsilon_4| = \frac{e^2}{2}.$$

Indeed, equal signs on the first two terms contribute mean absolute value 2, and opposite signs contribute mean absolute value e ; both cases have probability one half. All other tuples have zero defect by the balanced three-support equality. The probability of the exceptional four-type set is $24p_1p_3p_4p_5$. Equations (1.5) and (1.7) follow, using $\Delta = 4A$. Enumerating the ten three-type sets gives (1.6).

For (1.8), the one-sample section vanishes because every sample is at most two-sparse. Among the two-sample sections, only the one omitting e_2 can have four nonzero cofactors; its two additional types must be 4, 5, in either order. Its fourth cofactor is e^2 , giving the formula.

Every coordinate projection has both axis support numbers equal to one. The projection onto the 12 plane preserves the first new inequality, and the projection onto the 23 plane preserves the second. Thus the 12|3 product has support surplus zero at z_4 and e at z_5 ; the 1|23 product has the opposite behavior. At the 13|2 product and the all-line product both new normals have surplus e . All axis normals have surplus zero. The Euclidean distances to the block unions are precisely the same: an unhandled pair has a tail of length e and all handled pairs have distance zero. This proves (1.9).

3. What this says about a coupled proof

Sharp square-root matching even with arbitrary block choices The mean-distance obstruction is stronger than the fixed-coordinate statement. It persists if the line and plane blocks, and the sampled basis, can be chosen freely. The following elementary finite-set lemma makes this precise.

Finite-set lemma. Let $d \geq 3$, $0 < e \leq 1/2$, and let $R^d = \text{direct_sum } W_j$, with $\dim W_j$ equal to one or two. For

$$Z_e = \{e_1, \dots, e_d, e_1 + ee_2, e_2 + ee_3\}$$

one has

$$\max_{z \in Z_e} \text{dist}(z, \bigcup_j W_j) \geq \frac{e}{6\sqrt{d}}. \quad (3.2)$$

Proof. Suppose the maximum is $r < e/(6\sqrt{d})$. Assign each axis vector to a nearest block and choose w_i in that block with $|w_i - e_i| \leq r$. Let T have columns w_i . Then $\|T - I\| \leq \sqrt{d} r =: \epsilon < 1/12$, so T is invertible. The number of assigned axis indices in each block is at most its dimension. Summing the numbers and dimensions gives d on both sides; therefore the assigned index sets J_j partition the axes, have sizes one or two, and $W_j = T \text{ span}\{e_i : i \in J_j\}$.

For y in W_j ,

$$|\text{proj}_{J_j^c} y| \leq \frac{\eta}{1-\eta} |y|.$$

Take y to be the orthogonal projection of an arbitrary z onto W_j . Since $|y| \leq |z|$, the triangle inequality gives

$$\text{dist}(z, W_j) \geq \text{dist}(z, \text{span } J_j) - \frac{\eta}{1-\eta} |z|.$$

The two edges 12 and 23 cannot both be contained in blocks of a partition whose block sizes are at most two. For the corresponding uncontained edge vector z , its distance to every coordinate block is at least e . Since $|z| \leq \sqrt{5/4}$, its distance to every actual block is at least $e - 3\sqrt{d}$, a contradiction. This proves (3.2).

For every dimension, take the actual cone body

$$K_{e,d} = Q_e \times [-1, 1]^{d-3}. \quad (3.3)$$

Its cone law has the five three-dimensional direction-pair masses $(3/d)p_i$, and mass $1/d$ on each additional coordinate-axis pair. All $d+2$ masses are at least $3/(8d)$. Applying (3.2) therefore yields

$$\inf_{(W_j)} \mathbb{E}_{\nu_{K_{e,d}}} \text{dist}(X, \bigcup_j W_j) \geq \frac{e}{16d^{3/2}}. \quad (3.4)$$

The determinant and fourth-cofactor expectations are exactly

$$A_d = \frac{d! 27}{6d^d} A, \quad \mathbb{E}\phi_d = \frac{(d+1)! 81}{d^{d+1}} p_1 p_3 p_4 p_5 e^2 \sim \frac{(d+1)!}{8d^{d+1}} e^2. \quad (3.5)$$

For the first formula, a determinant needs one sample on each additional coordinate axis and three independent samples in the first three-dimensional subspace. For a positive fourth cofactor, a $(d+1)$ -tuple needs one sample on every additional axis and the four exceptional types $\{1, 3, 4, 5\}$ in that subspace. Its four nonzero cofactors have magnitudes $(1, 1, e, e^2)$, with all other cofactors zero. This proves (3.5) by ordered-type counting. The scalar deficit is

$$\delta(K_{e,d}) = (3/d)\delta_e \sim e^2/(16d), \quad (3.6)$$

either by the dimension-weighted product identity or by the same exact defect count. The section cost at the axis basis is $18 p_4 p_5 e^2/d^2$.

Scaling by $c = \sqrt{1+e^2}$ puts the entire cone law in the unit ball. Every basis matrix formed from its samples has $\|U\| \leq \sqrt{d}$. If its coordinate blocks are W'_j , their original blocks are $U W'_j$, and

$$\text{dist}(U^{-1}X, \bigcup_j W'_j) \geq \frac{1}{\sqrt{d}} \text{dist}(X, \bigcup_j U W'_j).$$

Consequently (3.4) gives a coordinate mean lower bound $e/(16 c d^2)$ for **every** invertible sampled basis and every line/plane partition. The bounded-law determinant mass has an explicit uniform lower bound: using $p_i \geq 1/8$ and $A \geq 6p_1 p_2 p_3 = 3/256$, one obtains

$$A_d/c^d \geq \frac{27d!}{512d^d(5/4)^{d/2}}. \quad (3.7)$$

It follows that, in every fixed dimension $d \geq 3$, no exponent greater than $1/2$ can replace the square-root exponent in a mean-distance matching lemma under its actual hypotheses, even when the law is required to be the cone law of a uniformly nondegenerate symmetric body and the block decomposition can be chosen freely.

The discrepancy is also linear in distance to the entire product class The same family satisfies a full-class geometric estimate:

$$\boxed{\frac{e}{120\sqrt{d}} \leq D(K_{e,d}, \mathcal{E}_d) - 1 \leq \frac{e}{1-e} \leq 2e.} \quad (3.8)$$

This does not improve the existing $1/d$ obstruction, since here the scalar deficit is of order e^2 . It shows, however, that the matching square-root loss can reflect an actual first-order product discrepancy, rather than only a badly chosen coordinate presentation.

We use a general elementary flat-facet observation. Suppose a full-dimensional body K has a facet with unit normal $\mathbf{n}$, support number $h > 0$, and a Euclidean tangent disk of radius ρ contained in the facet. If $K \subset E \subset (1+s)K$, the upper boundary of E , over that disk and in direction $\mathbf{n}$, is the graph of a finite concave function with height in $[0, sh]$ above the facet plane. At a differentiability point in the concentric disk of radius $\rho/2$, its gradient has norm at most $2sh/\rho$: restrict the concave function to each tangent line, and compare its derivative with secants to points at distance $\rho/2$ on either side. Such differentiability points exist by local Lipschitz continuity of finite concave functions. Their outward unit normal $\mathbf{n}_E$ satisfies

$$\text{dist}(\mathbf{n}, \text{span } \mathbf{n}_E) \leq \frac{2sh}{\rho}. \quad (3.9)$$

If E is an affine product of line and plane factors, every regular outward normal of E lies in one of its dual blocks, a direct-sum partition into subspaces of dimensions one and two. Indeed, if two primal factors were on their boundaries at a point, their nonzero normal cones would give at least two independent normal rays; that point could not be regular. At a regular point exactly one factor is on its boundary, and its normal annihilates all the other primal blocks.

For Q_e , tangent disks of radius $\rho=1/10$ lie in its five positive facets, with centers

$$(1, -1/2, 0), \quad (0, 1, -1/2), \quad (0, 0, 1), \quad (1 - e/2, 1/2, 0), \quad (0, 1 - e/2, 1/2).$$

For the two new facets the disk is taken in their actual tangent planes. All disks satisfy the remaining inequalities with room to spare for $e \leq 1/2$. In $K_{\{e,d\}}$, use the same centers with zero extra coordinates; the radius- $1/10$ disks in the full $(d-1)$ -dimensional tangent planes remain inside the corresponding facets. Each additional coordinate facet also contains such a disk centered at its coordinate-axis endpoint. All facets in question have the scaled normals in Z_e , namely $\mathbf{z}=\mathbf{n}/h$.

For any E in E_d with $K_{\{e,d\}} \subset E \subset (1+s)K_{\{e,d\}}$, (3.9) therefore puts every $\mathbf{z}$ in Z_e within

$$(1/h)(2sh/\rho) = 20s$$

of the union of E 's dual line/plane blocks. Equation (3.2) forces $20s \geq e/(6 \sqrt{d})$. Every Banach–Mazur sandwich can be rescaled to this form, and taking its infimum proves the lower bound in (3.8), without any attainment assumption. Finally $(1-e)[-1,1]^d \subset K_{\{e,d\}} \subset [-1,1]^d$ gives the upper bound.

A joint geometric gate remains necessary The family has deficit of order e^2 and a ridge-scale geometric discrepancy of order e . It does not threaten the conjectured optimal dimension-three power $1/3$, since e is smaller than a constant times $\delta^{(1/3)}$. Instead it shows why it is unsafe to strengthen the final volume gate by inserting a square into the mean-distance estimate. The small determinant factor e^2 is compensated by geometry spread over a ridge, whose cap-volume scale has exponent two rather than three.

By contrast, the three-coordinate corner truncation has $E \phi$ of order t^2 , δ of order t^3 , and support surplus of order t^3 . In that case the half-mass gate and a support-Lipschitz estimate lose a factor which the actual support cost already contains. The two families show different gates becoming active. The separate extreme bounds cannot be assumed to saturate on one body.

There is a useful exact weighted estimate. For the normalized cofactor tuple, let $g=1/2-t_1$ and $\phi=M t_4$ on $M>0$, and define $g=\phi=0$ on $M=0$. Since $t_4 \leq 1/4$ and $g \leq 1/2$ on $M>0$, Lemma 3.1 of the completed proof implies

$$t_4 g \leq \min(t_4/2, g/4) \leq \mathcal{D}(t), \quad \boxed{\mathbb{E}[\phi g] \leq N A \delta.} \quad (3.1)$$

This carries no power loss, but it cannot replace the omitted-basis matching cost by itself. Matching consistency must still be proved using a geometric cost which treats a shallow long circuit and two overlapping nearly-axis pair directions differently.

An unproved route toward the optimal $1/d$ is to produce a geometric witness at a product-containment defect t , classify the support involved by its active block dimension, and show that the cone-law masses times the circuit defect are at least $c_d t^r$ for some $r \leq d$. A proof must account for determinant factors, the normal-angle factors in localized facet mass, and non-polyhedral contact patches together. Merely optimizing the independent mean estimates does not establish such a statement. No complete lemma of this form is proved in this note.

4. Exact finite verification

`check_global_slanted.py` replays the cone-mass identity, the determinant expectation, all 625 ordered direction-type four-tuples with their 16 sign averages, all simultaneous two-sample sections, and the four block costs, at eight rational values of e . It also checks the product determinant, fourth-cofactor and scalar-deficit formulas in 32 cases across dimensions three through six. It uses exact rational arithmetic and exceptions rather than Python assertions. The finite checks support the displayed identities; the all-parameter proof is the algebra above.

The finite-set lemma, facet disks, and flat-facet geometric argument in Section 3 also received a separate analytical audit, with no gap found. This is model-assisted mathematical review, not external peer review or formal certification; the exact tests are not used to establish these geometric arguments.